ARA from premises to papers: construction, search, and the limits of licensed inference

An anonymised methodology report · Version 5, concept-based revision · correspondence through 30 October 2026, 06:46:47 UTC · live checkpoints explicitly dated

Abstract

ARA separates a bank of licensed moves (L), a frozen theory assembled from source commitments (T), and proposed hypotheses or targets (H). The scaffold was **designed and directed by a human, and built largely by AI coding agents under that direction**. This report organizes its development and use around those boundaries. Construction chooses which literature can enter T; window failures, source stance and atomic formalization can prevent a useful inference even when a passage has been acquired. Forks preserve one freeze while varying target context or hypotheses. Bank checks and polarity search then distinguish an admissible step from an unclosed obligation. The central representation bottleneck was measured directly in the shared freeze: 141/160 commitments were paper-prefixed atoms, and only two of 147 symbols spanned papers through a single alignment bridge. With no shared vocabulary in the source commitments themselves, wider bank-licensed search mostly constructed Boolean packages rather than the intended cross-paper inference. Completed packed, unpacked and externally seeded tests produced 100 OPEN routes and no substantive CLOSED-derived contribution. ARA did propose near-T2 targets; the exact T2 formulation came from a separate literature dossier. Output gates exposed unstable audits, citation loss and unsupported prose. The available fallback PDF analyzes those outcomes; two native sign-argument drafts were refused. Gap2 stopped before freeze. Gap3, shared-catalogue/wider-alignment repair, disclosed hypothesis-premise support and proposal-mode writing remain pending; the coordinator requested a stop to the larger forward run after its truncated census. The latter approaches separate arguing for a new bridge principle from mechanically deriving a conditional result. Deterministic figures, recorded human decisions, source hashes, worker credits and scoped cost accounting make the evidence and its limits inspectable.

1. L, T and H: the architecture and responsibility boundary

The human chose the research area, supplied the bibliography, directed the scaffold, approved budgets and provider changes, and set acceptance and submission priorities. Coding agents implemented, reviewed and operated it. ARA generation used separate provider calls; isolation of those calls does not establish isolation of development workers. The operating rule prohibited the human from supplying original philosophical arguments, premises, distinctions or a personal view. Recording actions does not independently certify competition compliance. The human-action ledger now contains **38 timestamped entries**, plus earlier untimed design and bibliography preparation. Worker use of a native `HUMAN` input boundary does not count as a distinct human intervention. [R1–R3, R34]

Every later effort is an instance of the same process: represent admissible evidence in T; propose something in H; try to license a move through L; record what remains open; and gate any written account on its actual support. Proposal-mode writing and proposed hypothesis premises extend this process in different ways. Neither licenses quietly converting a new idea into source evidence.

ARA separates **L**, the versioned bank of permitted move forms; **T**, the working table of source-linked commitments; and **H**, model-proposed cuts or hypotheses. A model is not allowed to create a license by asserting that a step is plausible. A step must instantiate an existing bank entry. Deductive entries require both shape checks and Z3 entailment. Defeasible entries require the relevant scheme and its critical questions; a failed or unsupported critical question leaves an obligation open. [R3]

Rich extraction records are immutable source evidence. A construction lifecycle selects a bounded working subset, formalizes it, records source stance and omitted support, and then freezes T. Search uses that freeze. A verified fork copies the frozen prefix and provenance, proposes new targets, and uses a separate cache. It reuses premises, not statistical independence. Reported or attacked source positions are not silently converted into endorsed premises. [R3, R4]

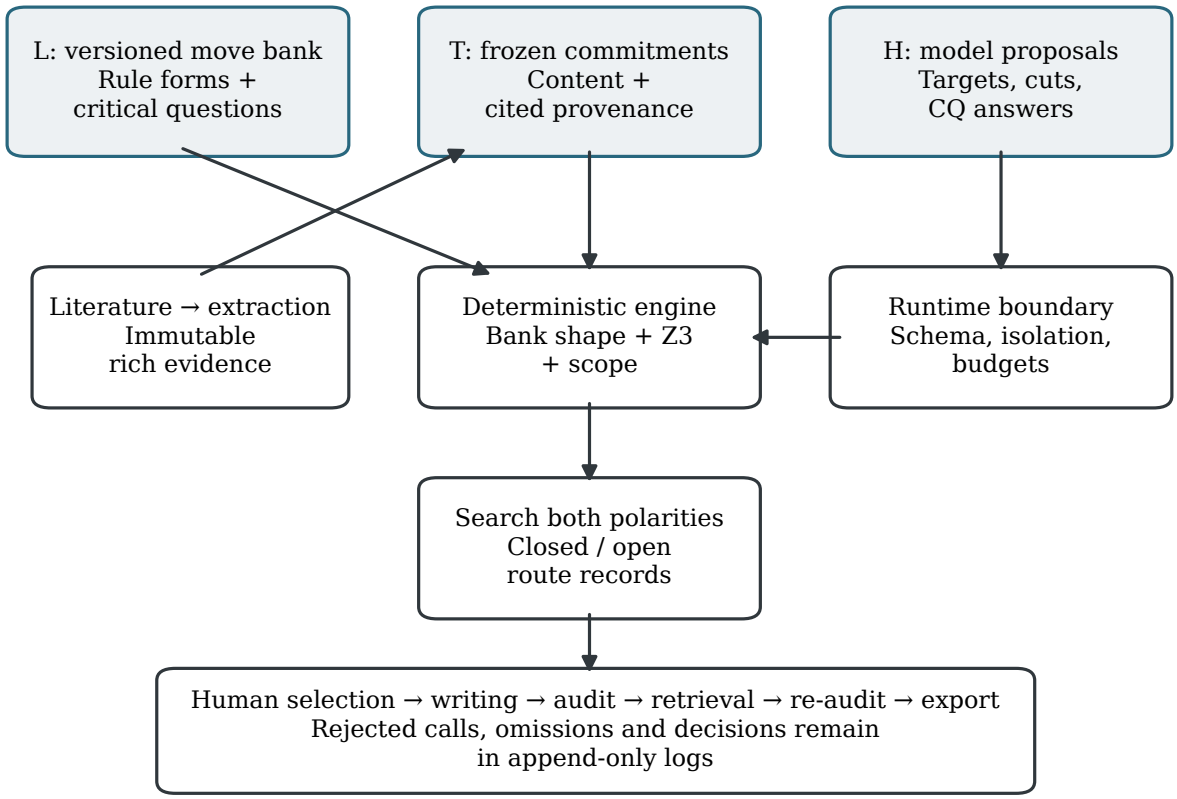
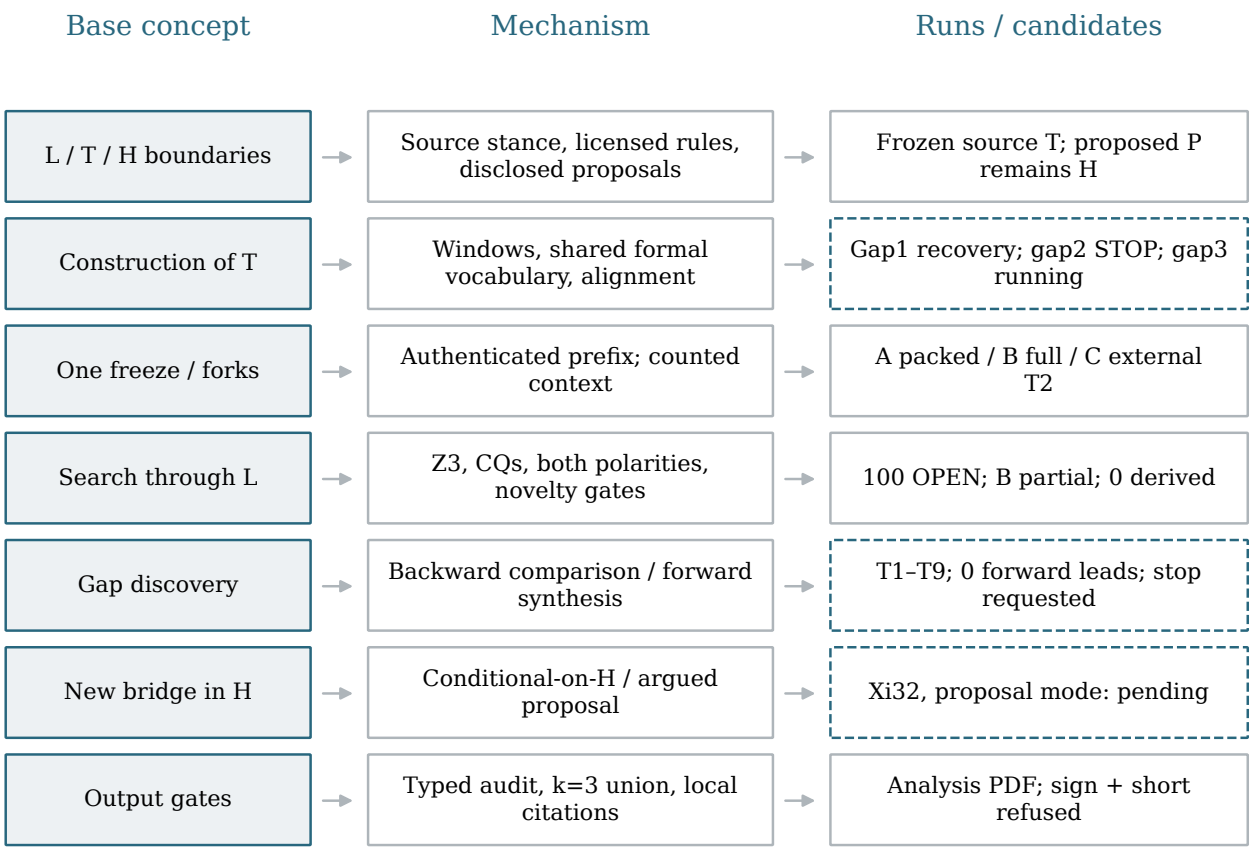


Figure 1. ARA separates the source of a license (L), the premises available to a run (T), and model-suggested hypotheses or cuts (H). Models propose; the deterministic engine decides formal admissibility. The final export path remains gated.

For each target, ARA searches both the target and its negation. The engine records proposed, licensed, discarded, and unresolved moves. A closed route means the engine's formal obligations were met under that version's policy. It does not establish that a source was faithfully translated, its premises are true, the result is new to the literature, or its prose is good philosophy.

Originality was pursued through several independent checks: avoiding direct restatement of T, testing single-support entailment and tautologies, classifying whether the target states an own thesis, checking the relation between its text and formal definition, requiring distinct source supports, and later comparing with retrieved literature. The gate design changed substantially. Its defects are part of the method's history, not evidence of an invariant guarantee. [R9–R11]



Dashed cells include pending work; arrows describe responsibility, not proven success.

Figure 2. One architecture organizes the experiments and extensions. Source construction determines T; L licenses search; H includes targets and, in a pending extension, disclosed proposed premises. Arguing for a bridge principle differs from deriving a result conditional on it. Pending mechanisms receive no success count.

This revision replaces the earlier chronological addendum with a concept-based account. Completed results are tied to their run or reviewed head. A live checkpoint is not a final result, a code review is not a philosophical validation, and a rendered PDF is not a native export. The companion data preserves earlier snapshots and records this revision's sources separately. V1–v4 and the earlier v5 checkpoint remain unchanged. [R34]

2. Constructing T: retrieval, windows, stance and alignment

2.1 Literature and extracted evidence

The initial bibliography contained **53 works**, dated **1967–2024**. Only **four** were from **2020 onward**, and none was later than **2024**. The first acquisition pass obtained **26 PDFs**, leaving **27** originals requiring access elsewhere. The accessible subset covered **1996–2024**, with **three** works from **2020 onward**. Intake accepted all bibliographic entries, but bibliographic intake was not full-text extraction. [R2, R5]

The first small-summary design was rejected because a **four-item, 400-character** cap could not preserve argument structure. Rich extraction instead retained page-linked premises, intermediate conclusions, inferences, distinctions, qualifications, and reported or attacked positions. The complete original-corpus pass produced **5,659 items** in **26 nonempty graphs**. Of those items, **363** were quarantined: **312 directly** and **51 because of**

dependencies. **5,296** remained eligible. **Nineteen** graphs were classified as extracted and **seven** as partial; **57** snapshot-scoped extraction gaps remained. These gaps concern source processing and must not be confused with open research problems. [R1, R12]

The later acquisition and currency audit added **28 full texts**: **25 modern selections** and **three** missing originals. The local collection therefore reached **54 PDFs**. The modern search recorded **44 candidates**; **19** modern selections and **24** originals remained unavailable, yielding **43 acquisition requests**. Access failure is a limit on the review, not evidence that a problem is open. Newly obtained texts were not part of the frozen pilot evidence at the supplement's checkpoint. [R5]

The small-summary cap was a design mistake against the stated aim of preserving arguments. Rich extraction removed that cap, but strict output contracts then rejected useful material. A cheaper Sonnet extraction produced malformed inference structure; a subsequent retry reduced a partial paper to **nine items**, compared with **108** from the Opus extraction. The cheaper path did not pass the richness test. The first complete-corpus attempts also stopped on reversed page ranges, missing inference dispositions, duplicate correction IDs, and reconciliation failures. [R1]

The repairs separated deterministic wire repair from invented source content, added bounded corrections, allowed disclosed reconstruction or a no-inference disposition, and retained accepted ranges through best-effort reconciliation. Shape validation alone had admitted placeholder-like junk: a later grounding pass quarantined **245 of 2,333** items in an intermediate corpus. Union publication repaired source loss across resumed snapshots. The final quarantine inventory remained visible rather than being deleted to improve a success count. [R1, R12]

OCR was tried on **82 affected pages**. Successive diagnostics reported **22** recovered pages under the first metric, then **15** under revised whole-page rules. Span-level repair recovered **131 of 677** damaged spans, with only **nine of 82** pages clean. These are different metrics, not a monotone improvement curve. Review found confidence dilution and required a further fix. OCR availability did not establish complete source recovery. [R1]

The later T1–T9 dossier obtained seven further lawful full texts and inspected the two modern comparisons that had previously been inaccessible. This is a dated acquisition update, not a retroactive expansion of any earlier freeze. Its substantive comparisons are discussed in §5. [R27]

2.2 Bounded selection and what a skipped window loses

The first rich build dropped **40 dependencies**, so no freeze was accepted. Later construction could not fit whole-paper provenance into a window: one singleton input was approximately **86,000 tokens** against a **70,000-token** batch limit. Filtering metadata to the selected window repaired that particular cause, but further refusals exposed the absence of a global plan. [R1]

The coordinator then justified a lower symbol-growth allowance using a claimed **70–90 tokens per commitment**. That claim was wrong. The pilot worker corrected the observed symbol cost to approximately **233 tokens per commitment**. The report retains the error and its correction. Bounded closure, disclosed support cuts, global working/catalogue budgets, and soft per-window shares replaced successive cap increases. An intermediate run had stopped at **802 tokens against an 800-token share**, despite remaining global capacity; the corrected policy allowed borrowing within the global bound. [R1]

A construction window is a counted request over selected grounded evidence, with dependency closure, formula and symbol checks, stance, page provenance and bounded corrections. A larger context allowance does not guarantee a usable reply. All-or-nothing window acceptance can discard many valid-looking items when one required inventory or definition fails. Skipped material must remain visible as an omission, not silently disappear from the account of T. [R16, R35]

2.3 Source structure and attribution

A resumed Codex pilot completed **20 searches** but returned only open routes. Diagnosis found **16 of 22** proposed steps with conclusion mismatch, a T of **166 standalone atoms**, and no target-symbol overlap. Structured source inferences, frozen-catalogue guidance, goal binding, joint slot formalization, and reference checks repaired concrete causes of empty search. The next pilot obtained closed routes, but they were depth-zero restatements of individual commitments. [R1, R10]

Own-thesis classification then became entangled with attribution: content definitions saying that an author held a claim made the classifier reject useful own-voice targets as reports. Content-only definitions plus separate stance/provenance repaired that representation. A subsequent lexical filter still rejected ordinary words such as “claims” or “states” in substantive content. Fork 2’s review identified **eight lexical-only exclusions**. The lexical screen became advisory; exact report frames remained hard failures. [R9]

These are representational checks. Endorsed, reported and attacked source positions remain distinct, even when each has a page citation. A model-alignment judgment is not an additional source paper. An atomic formula associated with a paragraph stating a rule is not itself an executable conditional. The later gap2/gap3 anchor results make that distinction concrete. [R28, R35, R36]

Measured bottleneck: paper-local atoms instead of shared predicates

The common explanation for failed cross-paper synthesis is the representation of T, not just the search or context budget. Direct inspection of `runs/codex-gap-1r/frozen-t.json` verifies **141 of 160 commitments as bare propositional atoms**, all with paper-prefixed names, for example `beddorgoldstein2021_info_sensitive_safety`. The remaining **19** have non-atomic syntax: 18 top-level implications and one conjunction. “Structured” here describes the formula tree, not proof that a paragraph's internal quantification, relations or normative force were faithfully formalized. All **147 catalogue symbols are zero-arity atoms**, not a shared predicate vocabulary. [R46]

Counting the papers attached to each symbol's formula occurrences through commitment provenance, **only two of 147 symbols span at least two papers**, and **both gain that status through the single alignment bridge**. Excluding the bridge, **zero symbols are reused across source papers**. The two are the MacFarlane assessment-sensitivity and Schaffer truth-perspectivalism atoms; their bridge remains a model judgment, not source endorsement. Thus the 2/147 count must not be read as two independently established equivalences. The completed alignment examined **150 candidate pairs: 149 unrelated and one directional entailment**; no catalogue symbol mapping changed. [R18, R46]

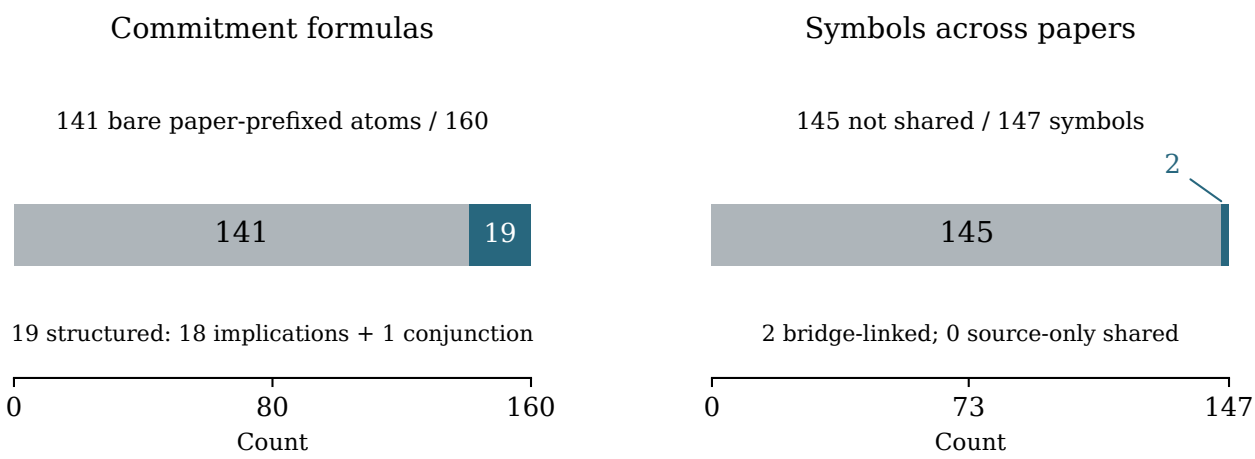


Figure 3. The same gap1r freeze underlies A/B/C and the forward sweep. Structured means non-atomic syntax, not established faithful formalization. Every catalogue symbol is a zero-arity atom. Cross-paper sharing counts formula occurrences using their source provenance; only the two symbols in the alignment bridge span papers. Excluding that bridge, source-only sharing is zero. Alignment tested 150 pairs (149 unrelated, one directional entailment) and merged no symbols. This quantifies the missing common formal vocabulary; it is not an impossibility theorem.

The bank cannot infer the internal content of an opaque proposition from its English name. In particular, an atom naming information-sensitive safety does not expose a reusable safety predicate, its arguments, or a conditional connecting the same assertion and contexts to retraction. The 19 non-atomic formulas and one disputed bridge did not supply the required cross-paper chain. Boolean conjunction/disjunction rules can still fire across otherwise disconnected atoms—as the forward census shows—but this packages premises rather than producing the intended substantive bridge. **No substantive cross-paper derivation was established in the recorded tests.** This is an evidence-based explanation of those outcomes, not a theorem that every possible bank application or future representation must fail. [R28, R44, R46]

Sigma r17 is pending at the 06:46:47 UTC instruction checkpoint. The urgent repair investigates structured formalization against a shared run-level predicate/term catalogue and a much wider, budgeted alignment pool. It must preserve source stance and provenance, disclose model alignment, and avoid inventing equivalences. Replay metrics should compare atom share, cross-paper symbol reuse and candidate licensed cross-paper applications. No repaired freeze, replay improvement or successful route is credited yet. Whether gap3's remaining alignment can adopt a wider pool without violating plan identity is itself pending; its running lifecycle is not silently changed. [R47]

2.4 Immutable identity and resumable alignment

The fresh gap pilot ran at detached revision `0a31afb`. It completed **74 construction windows: 53 accepted and 21 skipped**, retaining **159 working commitments from 22 of 26 papers**. These were pre-alignment artifacts. **There was no frozen T**, so the frozen-coverage criterion was not met even though the working set represented enough papers numerically. The supervisor exited with status **2** at **13:36:58 UTC on 29 October**, after **16,340.66 seconds**. [R16]

Alignment stopped at candidate **36 of 150**. It had accepted **35 judgments**: **34 unrelated** and **one right-to-left entailment**. The next candidate received **three invalid replies**, each changing the immutable symbol-ID fragment `r27_39` to `r27-39`. The responses called the relation unrelated, but their changed reference violated the exact-reference contract. **The validator correctly refused them.** The model text was not silently normalized and the working set was not promoted into a freeze. No bridges or canonical rewrites were applied; the final consistency phase was never reached. [R16]

The free `resume-construction --plan-only` check also refused: “construction resume requires a terminal window refusal.” Construction had already finished, and that revision had no alignment-resume entrypoint. Starting a new run with retained cache was the documented alternative; it would replay construction and could make new calls. That alternative was not executed as an in-place recovery. **Targets, forward synthesis, routes, writing, and an L3 resolution were never reached.** A rendering pass over an empty route inventory is not evidence of route-summary quality. [R16]

Coverage and semantic limits survived the completed construction stage. Skipped windows included capacity/empty-closure losses, a no-root selection, lexical attribution false positives, and provenance/schema refusals. The sole accepted entailment judgment was disputed on source review and involved an attacked description, not an endorsed safety–retraction bridge. Several original L3 anchors did not survive construction; a surviving retraction-related statement could not silently replace the full retraction rule. Four conflict inputs also lacked bindings because their required source contributed no commitment. These were qualified draft inputs, never admitted frozen gaps. [R16]

Independent review of this exact revision also retained blocking **INTEG-F1**: irrelevant fresh-atom wrappers could package existing cross-paper facts and receive a native derived label. This defect was independent of the earlier identifier stop. The pilot generated no target or route on which to test it; an empty padding-review inventory does not clear the defect. L3 therefore remains partially anticipated, with no resolution established by this pilot. [R16]

The identifier validator was right to refuse gap1's altered reference. The avoidable design error was asking a model to echo an immutable ID, letting one item's bounded failure abort the run, and offering no continuation from finished construction. Sigma r12 moved identity binding into the host-owned candidate record, requested relation/reason only, recorded per-item unresolved outcomes under a declared threshold, and added alignment continuation. Gap1r exercised that continuation with zero unresolved outcomes. Recovery preserves stage receipts; it does not certify the semantics of the retained bridge. [R17, R18, R21]

At `4cefe35`, recovery restored all **74 construction windows**, **159 source commitments**, and **35 accepted legacy judgments**. It completed **115 pending pairs** without replaying construction: **150/150 accepted judgments**, comprising **149 unrelated** and **one right-to-left entailment**, with **zero unresolved**. Applying the one model-alignment bridge produced **frozen T = 160 (159 source + 1 bridge)** across **22/26 papers**, with no canonical rewrites. This is executed continuation evidence, not merely a repair brief. The source inventory retained **134 endorsed, 10 reported, and 15 attacked** commitments. The inherited bridge remains semantically disputed; accepting its receipt did not establish a safety–retraction inference. [R18]

Sigma r12 supplied ID-free alignment replies and resume-alignment; later review closed its integration fixture issue. The design lesson is to bind immutable IDs in deterministic code, isolate a bad item's failure, preserve accepted judgments, and resume from completed stages. The recovered freeze has **160 commitments: 159 source entries from 22/26 papers and one disputed model-alignment bridge**. Zero unresolved alignments does not establish that the bridge is substantively justified. The literal all-source-stance criterion is **159/160, FAIL**, while paper coverage passes. Construction retained **53/74 windows**; its **21 skips** comprise 16 closure-bound failures, four formalization failures and one no-root window. [R18, R21, R28]

2.5 Gap2: more admitted context, less retained evidence

At the **30 October 03:51:27 UTC** pre-freeze check, gap2 had accepted **3/13 windows**, skipped **10/13**, and retained **57 source commitments from three papers**: MacFarlane 42, Schaffer 10 and Veltman five. Its stances were 46 endorsed, eight attacked and three reported. The correspondence records the stop decision at **04:04:01**;

later gap3 status confirms **gap2 stopped without a freeze**. The pre-freeze receipt itself still records a pending stop question, so the later stop must not be attributed to that earlier receipt. [R1, R35, R36]

All 26 papers remained in the counted plan. No selected closure was cut, and there were **zero context/budget/provider-refused windows**. The ten skips were terminal validation/formalization failures, not an extra category to add to the ten skipped windows. Increasing batch limits changed partitioning from 74 to 13 windows; copied cache entries could no longer provide exact-input reuse. Whole-window failures then lost more material. This observed granularity effect does not establish a controlled causal claim about model quality. Raising limits again would not address the actual validation failures. [R35]

Exact-ID comparison found **five retained, 52 added and 154 lost source rows** relative to the old 159. Only eight of 48 evidence IDs in one formerly skipped window were recovered; no old skipped window was fully recovered. Gap2 gained MacFarlane's general Retraction Rule and four qualifications, but represented them as five atoms. Both Beddor–Goldstein windows failed, so the earlier safety/worldly-information premises disappeared. Eight other connected conditionals did survive, but none supplied the missing safety-to-retraction or necessity/exhaustiveness bridge. Thus gap2 was worse for the intended L3 test without being a strict subset or worse in every respect. [R35]

One rejected definition described Joey as maintaining a modal belief; this was a possible false author-attribution hit. Another diagnostic normalized nullable wire fields before applying correction patches: that repaired a schema issue but still left **94/95** required commitments. Neither diagnostic justified admitting the rejected window. Dinges and several newer comparison texts remained absent from native extraction; acquisition or an anchor nomination cannot manufacture their premises. [R35]

2.6 Gap3 and construction repair: pending, with preserved checkpoints

Gap3 runs a **fresh lifecycle on reviewed `f76c1b3`**, with exact L3 anchors and the older absolute construction limits: 70,000 batch / 100,000 working / 60,000 catalogue tokens, two corrections, and a run-level 600,000 context setting. Its actual plan is **77 windows / 147 slots / cap six**. An earlier preview had predicted 74 windows and 154 slots; the second startup instead produced cap three, below gap1's cap four. It was stopped before source generation and preserved. A corrected multiplier and a guard verified actual cap six before generation. The first startup had separately refused a missing brief path before any model call. These were disclosed recovery steps, not silently uninterrupted execution. [R36]

Pending at 06:22:42 UTC: 20 of 77 windows completed, **13 accepted / seven skipped**, with **70 working commitments from nine papers**. All 11 nominated anchors were present. These are in-progress working counts, not a freeze or completed coverage result. The pre-freeze guard requires at least 20 papers and endorsed Beddor–Goldstein safety/worldly-information anchors; alignment, the single freeze, source/formula review, L3 binding and target/search results remain pending. [R36]

The two accepted anchor windows preserve six MacFarlane roots and five Beddor–Goldstein roots, including the missing safety side. But **all 11 formulas are atomic**, with no connected conditional from those windows. MacFarlane's q4 preserves the permission/commission distinctions, q5 separates permissible rejection from obligatory retraction, and attacked n35 remains attacked. The retraction illustration remains reported. The Mighty Failure paraphrase substitutes a credence formulation for thinking that one might have passed; that change must not be treated as an established equivalence. Improved inclusion does not yet make the rule executable or establish faithful formalization. [R36]

Sigma r16 is **pending at this revision's 06:28:53 UTC instruction checkpoint**, with no final completion or review credited. Its requested native mechanisms are dependency-safe partial acceptance, one bounded split retry, corrected attribution false positives, and safe parallel windows. Flags are to default off; rejected items and their dependants must remain excluded. These changes must not alter the running gap3 lifecycle. A reviewed later lifecycle may opt in; the operator's later parallel-work instruction supersedes the earlier scheduling-only framing. The ~09:30 review target and ~10:00 gap3 freeze check are plans, not completion times. [R34, R37]

Later report snapshot, 30 October 06:44:42 UTC: sigma16 describes local partial acceptance, one split retry and a precise construction attribution screen, but final full-suite/control/committed-file receipts and independent review remain pending. The focused recovery suite reports 40/40. A whole-response complexity bypass was discovered and fixed during implementation; the first full-suite run was interrupted for that fix and is not final proof. Dependency/symbol pruning and whole-response bounds remain required. **Parallel windows are deferred**, with no concurrency flag or no-op path: safe concurrency still needs deterministic snapshots, reservations, collision handling, ordered publication and crash/resume proof. Split children also require separate retry-entry versus primary-window coverage counts. Combined integration with hypothesis-premise work has not been tested at this checkpoint. These are author-reported local results, not a reviewed gap4 readiness verdict. [R45]

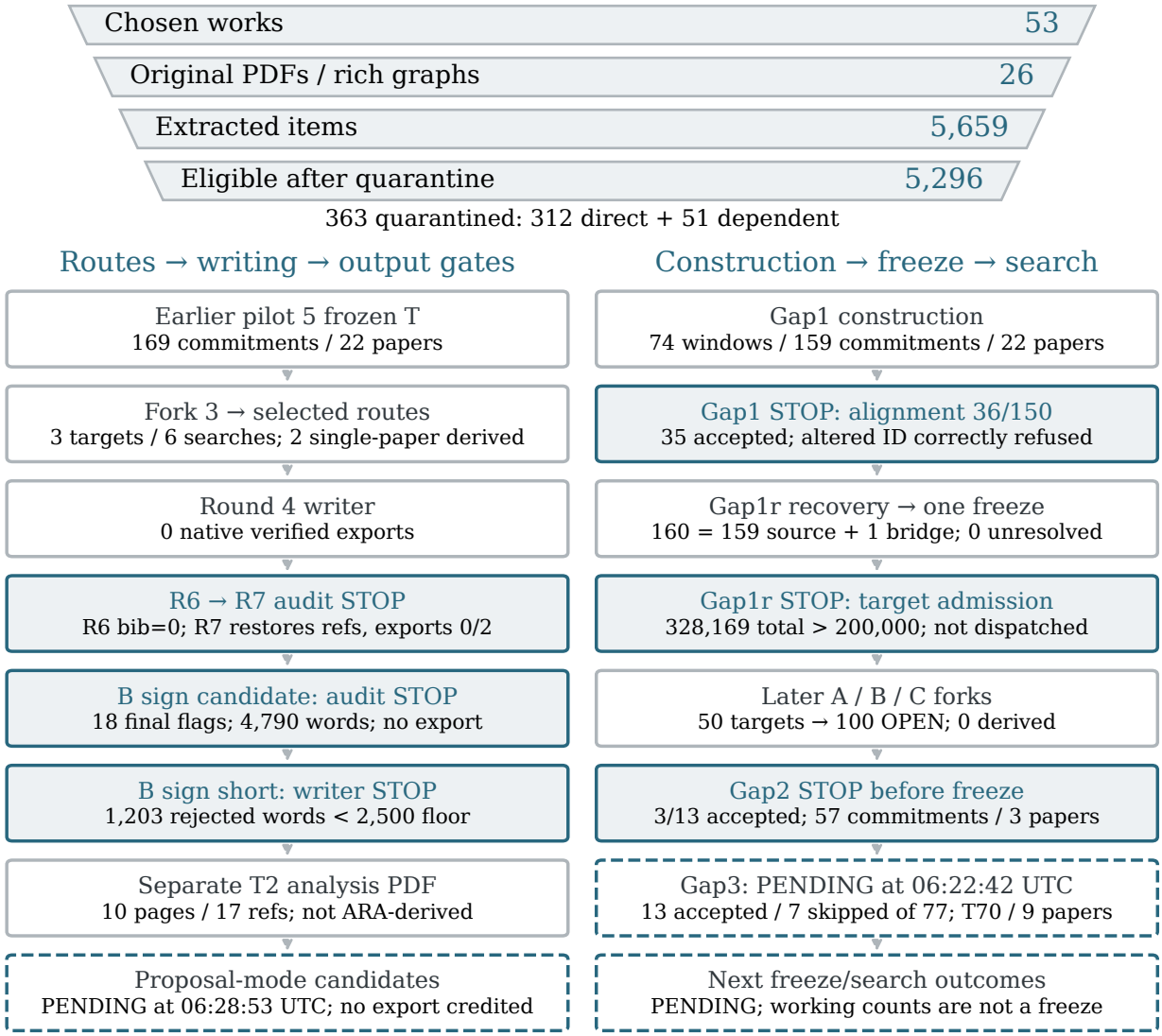


Figure 4. Shared historical extraction feeds distinct lifecycles and output attempts. Gap1 stopped at alignment; recovery froze T but stopped at target admission. A/B/C later completed without derived closure. Gap2 stopped before freeze; gap3 counts are live at 30 October 06:22:42 UTC. R7 and both sign drafts refused. The separate analysis PDF is not a native export. Proposal-mode outputs remain pending at 06:28:53 UTC.

3. One freeze, controlled forks, and context admission

3.1 Freezing a theory is an evidence boundary

A lifecycle constructs and aligns T, then freezes once. Fork-from-freeze reuses that authenticated theory and its provenance; it does not import improved evidence retrospectively or make the experiments independent. Gap1r reused completed construction and accepted alignment judgments. A/B/C then reused its byte-identical freeze. Gap2's unaccepted rows never entered such a freeze; gap3 starts a new construction lifecycle. [R18, R28, R35, R36]

3.2 Client windows and the refused target request

The first L3 target request was then refused **before dispatch**: **311,785 estimated input + 16,384 output allowance = 328,169**, against a **200,000** bound. Thus “312k > 200k” describes the input estimate alone; admission counted the larger total. **No target call, target, route, selection, or L3 resolution occurred.** A prior readiness check estimated only **51,998 input tokens** and did not represent the full target request. The full prompt was **627,750 UTF-8 bytes**, including **332,316 bytes** of provenance. The stopped request incurred no generation spend. Recovery's supervisor elapsed time was **13,773.98 seconds**, ending with status **2**. [R18]

Pinned Codex client metadata advertised a **272,000 default / 872,000 maximum** context window. These are client settings, not verified service capacity. The adapter initially failed to pass `model_context_window`. After the operator authorized loosening the token limit, sigma r13 forwarded it explicitly and added gap-scoped packing. The offline one-hop plan retained three full and 157 compact rows and estimated **122,108** total tokens. It was a plan, distinct from the subsequent actual two-hop fork A. [R19]

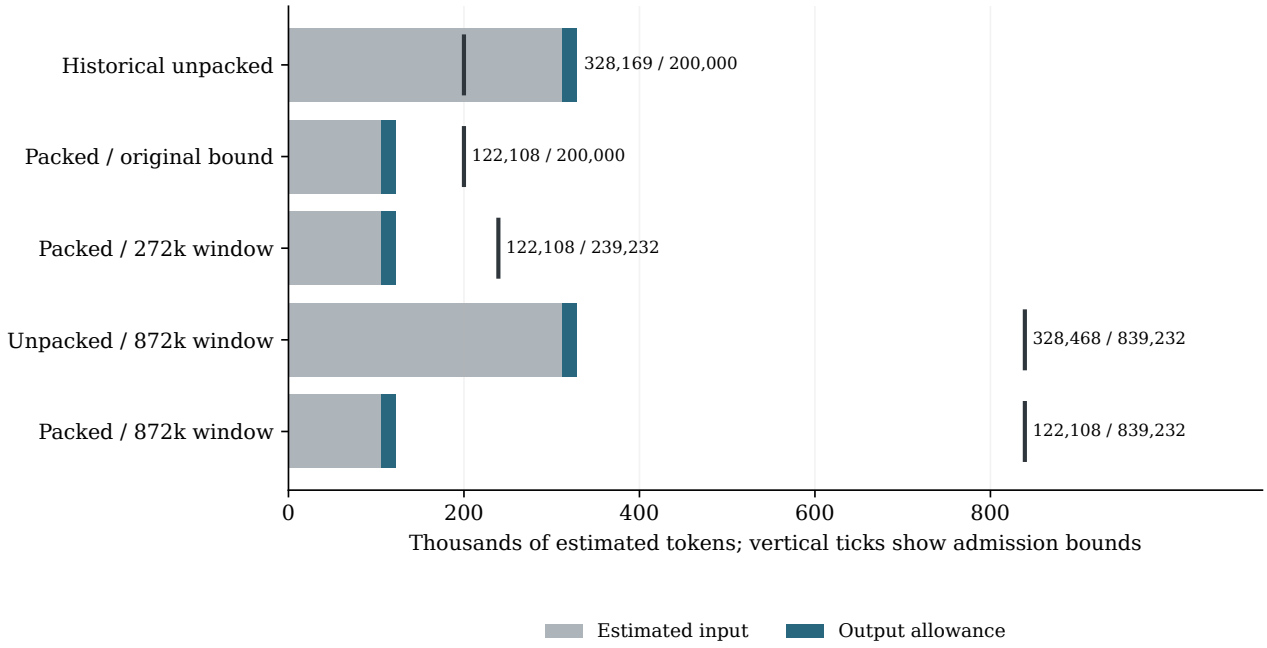


Figure 5. Full L3 request totals include output allowance. Only the first row is a historical refusal; the remaining rows are free offline plans. Packing uses one hop (3 full + 157 compact commitments); the later actual two-hop A results are reported separately in §3.3. This historical unpacked template adds 299 tokens. The later stage-window defect was closed by r14 review. These plan estimates do not prove service capacity.

Sigma r13's stage-specific configuration split could admit 587,839 estimated tokens under an effective 272,000 window because a larger inherited bound remained active. Sigma r14 normalized admission and provider configuration before cache lookup and explicitly refused null clearing. Kappa4 closed S13-F1/F2 at `7d19ccc`; the fix was merged into `893985a`. A/B/C used run-level windows only and avoided the historical trigger. These are offline enforcement findings, not proof of a model's maximum serving window. [R24, R30]

3.3 Actual packed and unpacked requests

A's actual two-hop L3 context retained **three full and 157 compact commitments**, including every frozen formula and stance. Its first live estimate was **105,722 input + 16,384 output allowance = 122,106** tokens; offline replay was 122,108. B retained full provenance for all 160 commitments: **312,084 + 16,384 = 328,468**

estimated tokens. A used a run-level 272,000 window and 239,232 bound; B/C used a run-level 600,000 window and 567,232 bound. No stage override triggered the historical stage-window defect. [R28]

Actual first-call usage was **60,702 input / 958 output / 3,968 cache-read** for A and **171,899 / 699 / 0** for B. The larger limit admitted a conservatively estimated request. These receipts do **not** show that the provider needed more than 200,000 actual input tokens or establish its maximum serving capacity. A/B also differed in ten semantic classifications before packing, despite identical pair inputs; token bounds differed, and there was only one arm of each type. Target-count differences cannot be causally attributed to packing. Their deterministic preparation was interrupted and explicitly recovered, with original receipts and freezes preserved; these were not uninterrupted runs. [R28]

The completed fork outcomes are in §4. Raising an admission limit admitted a conservatively estimated request; it did not supply missing premises, improve the freeze or establish a causal quality benefit. Fresh usage is reported separately in §8. [R28]

4. L licenses moves; polarity search exposes open obligations

4.1 Formal search and the outcome census

The operator-approved acceptable pilot required a clean native selection boundary, source coverage of at least **20 of 26 papers**, at least **three** genuinely derived routes, and readable summaries/reporting. Each qualifying route had to state a novel own thesis, avoid single-commitment entailment, use at least **two distinct supports from two papers**, respect stance, and contain a licensed inference. Pilot 5 met the clean-run, coverage, and rendering requirements but failed the substantive-route requirement. A native label alone did not satisfy the criterion. [R1, R6]

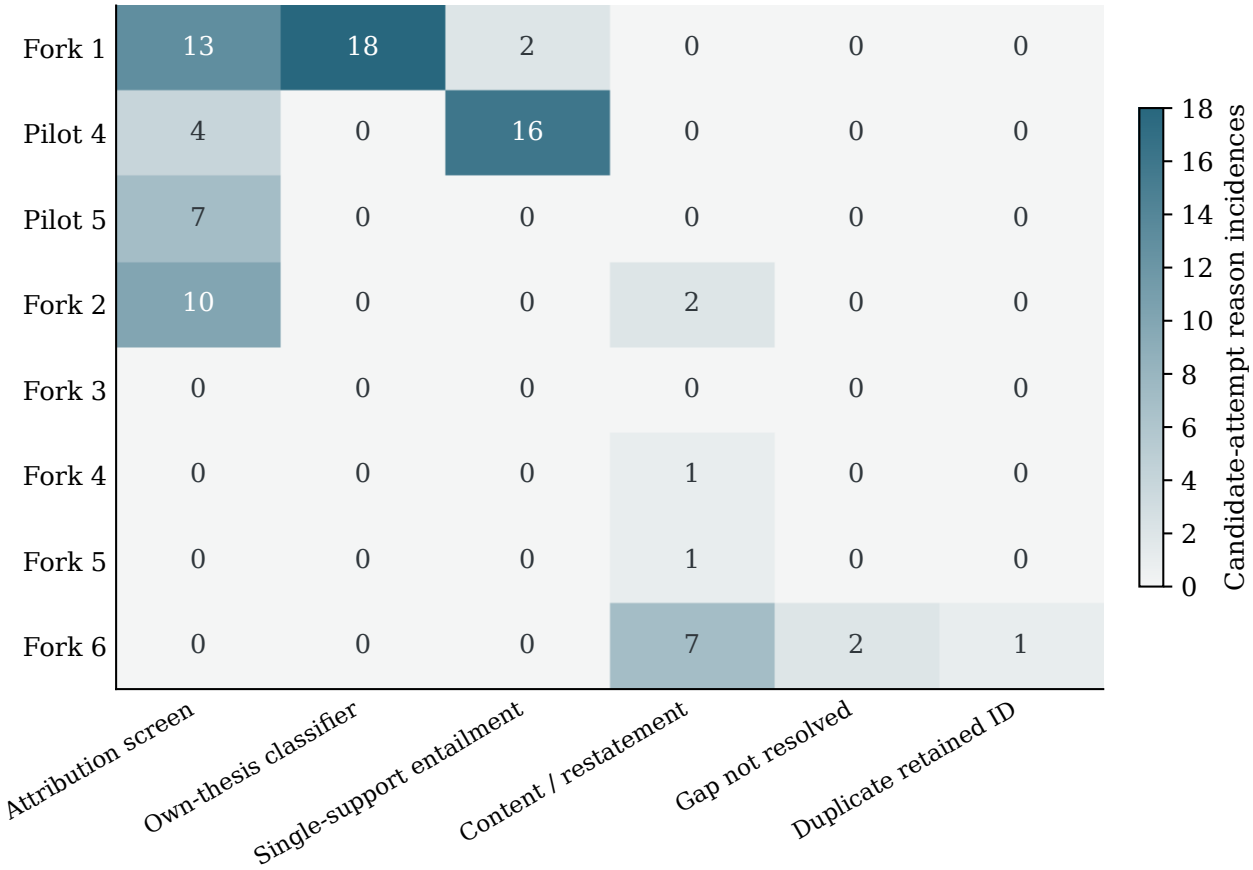


Figure 6. Native rejection reasons across pilot generations. The same candidate-attempt can occur in more than one column; columns must not be summed into a rejection total. “Attribution screen” includes lexical false positives, especially before that screen became advisory. Repeated proposal attempts count separately. Zero means no recorded rejection under that reason, not proof of semantic novelty.

The chart counts candidate-attempt incidences under the recorded gate reason. Reasons can overlap. In fork 1, an attribution screen and the own-thesis classifier could both reject a candidate; adding the columns would overcount rejected candidates. Later pool receipts record an explicit retained/rejected disposition. The figure preserves this difference rather than treating all generations as a controlled, identical experiment. Exact rows and event hashes are in the data supplement.

The gates caught several real defects: single-premise restatements, unused-antecedent conditionals, Boolean packaging, renamed assertions, ornamental proof steps, and unsupported gap-resolution labels. They also created false positives and exclusions: attribution vocabulary, an order-dependent paper count, stance-blind support slicing, a duplicate target identifier, and the outer target cap. Passing all gates was necessary under a given policy but never sufficient for literature novelty. [R4, R9–R11]

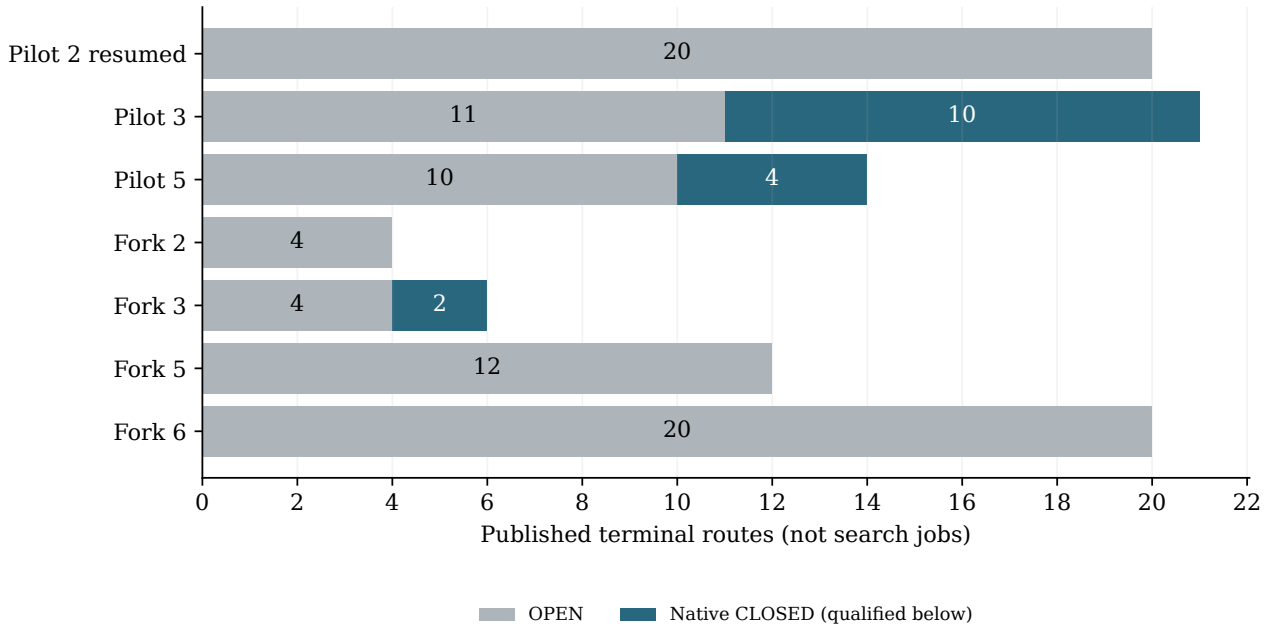


Figure 7. Closed is a formal outcome. Pilot 3's ten closed routes were trivial restatements; pilot 5's four were conjunction packaging; fork 3's two derived routes used a single paper. None met the complete accepted cross-paper criterion. Pilot 3 published twenty-one routes from twenty polarity searches.

The operator selected the two available fork-3 derived routes only after their single-paper limitation was reported. Those routes supplied writing experiments. The later L3 priority was a separate research direction, not a retroactive claim that the selected routes had already resolved that gap. No successful cross-paper pilot is established by the evidence covered here.

4.2 Novelty gates must allow synthesis without allowing padding

The whole-T novelty gate was another substantive design error. Requiring that the union of T not entail a target prevents deductive synthesis across papers, even when no single paper entails it. The corrected local policy allows union entailment and tests each paper's endorsed closure separately, while retaining anti-padding, stance, fidelity, and support checks. The placeholder regression combines P from one paper with P → Q from another and derives Q. This demonstrates the corrected mechanism, not a real literature result. The final bounded real-freeze census, at depth **two** and a **500-consequence cap**, found **zero qualifying candidates**. An earlier six-candidate census was superseded after a packaging defect was fixed. Cross-paper symbol alignment was still a separate integration boundary. [R11]

The R28 repairs distinguish a genuinely joint consequence from Boolean packaging and reject inconsistent endorsed-source unions rather than deriving arbitrary novelty from contradiction. Kappa4's later xi29 review closed R28-F1/F2 at the integrated head. Gap1r's previously observed packaging leads were rejected under that repair; its bounded formula union had been SAT, so the inconsistent-union trigger was absent in that freeze. A passing gate is still weaker than literature novelty or faithful source interpretation. [R18, R27]

4.3 Why A/B/C all remained OPEN

All three forks reached native `awaiting_selection`. No route had been selected at their handoff checkpoint. A/B ran on historical `c99c6b3`; C ran on reviewed `893985a`. All retained the same authenticated freeze: **160 commitments = 159 source entries from 22/26 papers + one model-alignment bridge**. The source-only

stance criterion remains **159/160, FAIL**; paper coverage passes. The disputed bridge is not source endorsement. Construction accepted **53/74 windows** and skipped **21**: 16 closure-bound failures, four formalization failures, and one without topical roots. Larger later contexts did not restore those omissions. [R28]

Fork	Targets	OPEN routes	CLOSED-derived	Demonstrated substantive contributions	Terminal argument content
A: packed, two hops	27	54	0	0	All 54 have a single OPEN root.
B: full provenance	20	40	0	0	39 single roots; one three-node partial sign argument, non-L3.
C: external T2 and directions	3	6	0	0	All six have a single OPEN root; no accepted inference, cut, support or CQ answer.

B's partial argument retains a sign inference and two source leaves; CQ1 is supported and CQ2 unsupported. One source's underlying inference is omitted. It remains OPEN and does not establish L3, but it does contain material that can be drafted and criticized. The all-candidates queue contains `gap1pu-B-sign`. C's exact biconditional and its two directions were disclosed external hypotheses from eps2's dossier, not ARA discoveries. Their formulas were whole-statement atoms; the biconditional and contextual quantification were not decomposed and the direction atoms were not formally linked. These are bounded unsuccessful tests, not countermodels or impossibility theorems. [R28, R33]

No closed safety–retraction argument follows from those experiments. B's partial sign argument is useful enough to attempt writing, but its unsupported CQ and omitted underlying source inference remain obligations. C is the focused test of externally supplied T2; its six root-only routes contain no licensed step. ARA's near-T2 proposals, the exact external formulation and the partial sign argument must not be collapsed into one result. [R28]

5. Finding gaps backward and forward

5.1 Backward: conflicts, missing connections, and literature currency

The initial gap detector found **three source-level open-problem flags**. It did not establish that those problems were still open in the current literature. The follow-up checked the source context, later literature, and proposed resolution theses. Its verdict was **one addressed, two partially addressed, and zero certified still open**. [R4, R5]

Flag	What the system treated as unresolved	What the currency check found
G1	Whether disagreement motivates a relativist account and where the effect belongs	Later theories and experiments address the comparison. Remaining disagreement does not certify an untouched problem.
G2	Nested probability and the interpretation of a particular reply pattern	Later formal/pragmatic models and experiments address the mechanism. The exact remaining scope was only partially settled by the accessible material.
G3	An older absence-of-model or absence-of-diagnostic claim	MacFarlane’s later models and fixed-use/retraction diagnostics address that claim. Relevant works were already in the original accessible corpus.

L3 emerged from the failed search for connecting premises and a targeted bridge-literature investigation: could worldly-information safety support an assessment-sensitive retraction obligation? The initial acquisition report classified the link as unknown. The deeper check revised it to **partial**. The safety anchor already discusses retraction and assessor information; Dinges's later work connects estimates of epistemic qualities, including safety, to retraction while allowing a rival semantic interpretation. These are substantive anticipations. The accessible texts did not establish the specific safety-to-retraction derivation sought by ARA. [R5]

The focused search covered **2018–2026** and retrieved indexed first-generation citation sets, including **11** works citing the safety anchor, **116** citing the earlier assessment-sensitive account, and **449** citing the later book. Those counts describe retrieved index records, not papers all read in full. The index omitted an actual book citation found in a read PDF, demonstrating that absence from a citation graph is unreliable negative evidence. At that earlier checkpoint two especially relevant modern comparisons remained inaccessible; the later dossier obtained and inspected both. [R27] [R5]

A separate live semantic-conflict pass classified **60 pairs**: **three contradictory**, **seven in tension**, **35 compatible**, and **15 unrelated**. Source/currency review of the ten flagged pairs yielded **nine partial** and **one unknown**, with none certified still open. L3 was omitted by that bounded top-pair retrieval and was investigated separately. The operator approved the restrained framing **"not found within stated search scope, partially anticipated."** That framing is retained here. It is not a claim that ARA resolved L3 or established priority. [R1, R14]

5.2 The dossier is a comparison map, not a proof

Eps2's completed dossier classifies **nine analyst-generated theses: four anticipated, four partially anticipated, one not found within scope**. Its only not-found thesis is the exact T2 necessity-and-sufficiency connection. The verdict does not establish truth, priority, feasibility or an exhaustive current gap. Seven acquired full texts support this analysis without becoming premises in the old freeze. [R27]

Dossier thesis	Comparison subject	Bounded verdict
T1	Safety failure sufficient for obligatory retraction	Partially anticipated
T2	Safety failure necessary and sufficient for obligatory retraction	Not found within scope
T3	Asymmetric loss without relativism	Anticipated
T4	Stable acceptance and withdrawal	Partially anticipated
T5	Assessment-relative worldly-information selector	Partially anticipated
T6	Flexible truth assessment, current-context retraction	Anticipated
T7	Retraction, cancellation and uptake	Anticipated
T8	Later responses help determine earlier content	Anticipated
T9	Fragmented/topic-sensitive knowledge with a retraction norm	Partially anticipated

ARA did propose near-T2 targets. Kept B/1 and B/2 partially match T2's necessity direction, while changing the subject or normative force: knowledge claims, warranted retraction, or an assessor requiring retraction are not the exact fixed-use, original-speaker obligation. Kept A/3 is only thematically related to T5. None is exact T2 or either exact direction. C's exact T2 came from eps2 and was supplied as a disclosed external test. Its

sufficiency probe is closest to partially anticipated T1; the global T2 not-found label cannot override that closest-thesis comparison. Negation routes inherit only a comparison label, not a positive thesis's literature support. [R28, R38]

5.3 Forward: what premise sets jointly imply

Backward investigation begins with a conflict, failed connection or proposed target and asks which premises would support it. Forward synthesis begins with available premise sets and asks what licensed consequences follow that no individual paper supplies. The original whole-T non-entailment gate obstructed precisely that synthesis; the repaired per-paper gate allows joint entailment while rejecting restatement, inconsistent support and packaging. [R11, R27]

The old native forward pass on gap1r was bounded at depth two / 500 consequences and returned two packaging leads, with **zero substantive contributions**. The answer to whether every premise combination and route had been searched is therefore no. The bounds and available bank schemes delimit coverage. [R18]

The **fwd sweep is pending at 06:28:53 UTC**. Its operator-approved brief directs the native forward path on the authenticated 160-commitment gap1r freeze at reviewed `f76c1b3`, with broader bounds and disclosed remaining truncation. Candidates must survive stance, SAT, licensed-inference, essential multi-paper support, per-paper novelty and anti-padding gates, then enter native target/polarity search. Literature verdicts would remain `pending-eps2` until checked. No completed sweep report, qualifying conclusion, route or queue count is credited here. This assignment explores premise combinations; it cannot repair missing evidence or justify the disputed alignment bridge by itself. [R39]

Later report snapshot, 30 October 06:44:42 UTC: the completed depth-six / 20,000-consequence census reports **zero surviving candidates**, **zero recorded support conflicts**, and **truncated=true**, taking 493.733 seconds. Retained consequences comprise 9,180 conjunction introductions, two conjunction eliminations and 10,818 disjunction introductions; an additional licensed disjunction hit the insertion cap. Only 11 non-scoped deductive bank entries have fresh forward instantiation; 23 defeasible and two scoped deductive entries need existing closed-route witnesses. Rule order can exhaust the cap before later rules or layers, so configured depth does not prove depth reached. At that snapshot the **50,000-consequence general target pool and subsequent search remained running/pending**; no candidate route or queue addition is credited. Its 30-target / three-attempt proposal limits remain. This is a completed bounded census inside an unfinished sweep, not an exhaustive absence result. [R44]

Connection to construction: the 20,000-consequence result is the operational signature of the sparse shared vocabulary measured in §2.3: 141/160 opaque commitment atoms, 147 zero-arity symbols, and only two cross-paper symbols supplied by the disputed bridge. The recorded conjunction/disjunction enumeration produced **zero leads**. A/B/C likewise ended **100 OPEN / zero CLOSED-derived**; changing packing or seeding exact T2 did not change their shared T. Together these observations identify inadequate shared formal structure as the unifying bottleneck. Enumeration order, bounded search, missing premises and source fidelity remain additional limits; the diagnosis is not a controlled ablation proving a single exclusive cause. At **06:46:15 UTC**, correspondence records the coordinator's instruction to **stop the wider forward sweep and prioritize sigma r17**. This is a stop request; no process-stop receipt or final 50,000-run result is credited in this report. [R1, R28, R44, R46, R47]

6. H as a proposed bridge: conditional derivation versus argued proposal

6.1 The missing work is substantive

The missing sufficiency connection links worldly-information safety failure to an obligation concerning the same earlier assertion with fixed use and specified assessment contexts. A necessary condition on knowledge does not itself create that obligation. The necessity direction additionally needs an exhaustive or excluding premise covering the relevant grounds of obligatory retraction; a sufficient trigger cannot simply be reversed. Permission to reject, actual withdrawal, a knowledge claim and an obligation to retract are distinct. Gaining MacFarlane's prose rule is relevant, but neither its atomic encoding nor a generic alignment judgment supplies all those connections. [R27, R28, R35, R36]

6.2 Hypothesis premises: a pending native extension

Xi r32 is **pending at 06:28:53 UTC**, not a completed derivation. The brief proposes admitting an explicitly disclosed bridge principle P through native construction before the single freeze, with `kind=hypothesis`, `stance=proposed`, a proposer, rationale and linked target. Normal formalization and consistency checks still apply. Hypotheses may not count as source papers or source support; a qualifying route must still have at least two source papers besides H, and simply restating P with padding must fail. Writer/audit roles must distinguish the paper's proposal from attributed source claims, and undisclosed H must prevent export. [R40]

The intended result is **“conditional on H”**: mechanically deriving a thesis from P plus the frozen source premises. This does not mechanically derive or establish P itself. A displayed “if P, then T2/B1/B2” result requires an actual native derivation and an adequate representation of the conditional; the assigned feature and proposed config entries are not evidence that either exists yet. P must not merely restate the target, smuggle in a source endorsement or satisfy the multi-paper requirement by being counted as another paper. Gap4 hypothesis-premise work is a planned parallel lifecycle and must not alter gap3. [R40]

6.3 Proposal-mode papers argue for P

Nu4's proposal mode is separately **pending at 06:28:53 UTC**. It is meant to let a paper state and defend a new bridge principle as its own proposal, with explicit reasoning, objections and the closest literature, while source claims retain citation requirements and unmarked unsupported claims fail. Proposed typed roles are `proposed_principle` and `argument_for_proposal`; k=3 union, bounded repair and local citation gates remain required. The assignments include exact T2, B/1, B/2 and A/3, followed by eligible new gap3/gap4 targets. These are assignments, not four completed papers or exports. [R41]

Required origin labels distinguish **“ARA-proposed thesis; bridge principle argued, not mechanically derived”** for the native target proposals from **“eps2-proposed thesis (tested in ARA); bridge principle argued, not mechanically derived”** for exact T2. Both disclose **proposal-mode: argued, not ARA-derived**. A later conditional-on-H route would add a distinct mechanical result without erasing the argued status of H. Pi3's polish must preserve these labels. The existing fallback analysis remains a third category: analysis of the output, not a proposal-mode or native derived export. [R34, R40, R41]

7. Writing and verification are gates on output

7.1 Typed claims and the early writing results

Repeated whole-draft retries did not supply actionable length feedback. Section budgets and append-only expansion preserved accepted material while filling deficits. Citation splitting, identity screening on resumed drafts, and revision-marker handling also required repair. Typed auditing improved the distinction between exposition and unsupported argument, but audit judgments varied markedly across runs. A zero-flag model judgment is evidence about that invocation, not an independent semantic proof. The retained adverse judgments and the final retrieval failures remain part of the record. [R7]

The initial whole-essay writer produced **eight attempts** across the selected routes, all below the then-configured **4,500-word** minimum; their range was **1,590–4,376 words**. No audit was reached. A sectioned writer then used an outline with section budgets, deterministic assembly, and targeted expansion. Its first length-successful draft reached **4,642 words**, but failed audit on **four sentences**. A repaired **4,604-word** draft passed a subsequent native audit, then stopped before external evidence retrieval. [R1, R7]

Later evidence changes the interpretation of that pass. A new binary audit flagged **101** sentences in the same previously accepted draft. Typed auditing distinguished source claims, argument steps, framing, objections, and unsupported claims. In the later recorded comparison, route 1 began with **75/245** flags and route 2 with **25/245**. Repairs yielded **4,714-word** and **4,619-word** drafts with zero flags in their final pre-verification audits. Both native runs nevertheless stopped at `search_evidence` with `cli_tool_event`. An isolated web diagnostic succeeded, but neither native lifecycle produced accepted retrieval evidence, citation/originality classification, the final audit, or an essay export. **Verified native exports remained zero at the round-4 checkpoint.** [R7]

Table 1. Round-4 typed pre-verification audit, reproduced from the writer's receipt. "Audit passed" does not mean that retrieval, citation/originality verification, or export passed. [R7]

Route / stage	Native words	Flags / sentences	Outcome
Route 1 imported draft	4,604	75/245	Revision required
Route 1 accepted repair 1	4,714	0/245	Pre-verification audit passed
Route 2 imported draft	4,638	25/245	Revision required
Route 2 accepted repair 1	4,525	74/244	Revision required
Route 2 accepted repair 2	4,619	0/244	Pre-verification audit passed

The increase from **25 to 74 flags** on route 2 shows why these labels should not be read as a stable quality score. Native word counts include machine markers. Excluding both node and citation markers, the final drafts contain **4,578** and **4,498 words** respectively; route 2 is two words below the then-configured **4,500-word** floor under that stricter count. Neither round-4 draft was a verified export. Nu4 r5 was pending at the version-2 checkpoint; its later completion and round-6 limitations are reported below. This historical table is unchanged. [R7, R17, R20]

A separate publication branch polished the preserved earlier route-1 draft through isolated Claude Opus calls. It produced a **4,743-word**, **12-page** PDF with **two deterministic figures**. Its native audit covered all route nodes and its exact T-citation membership check passed. External verification remained blocked on that branch. This polished artifact is evidence of successful prose and rendering work on one selected route; it is not the L3 gap paper and is not proof of a verified competition entry. [R8]

7.2 Audit nondeterminism, union gates and citation loss

Round 5 (`0368a17`) identified the refused event as `item.completed → web_search → action.type=other` and permitted that exact native web event only in `search_evidence`. Both routes then completed native export with originality labelled `possible_overlap`. That label is not a novelty pass. After quotation cleanup, route 1 received **44 flags**, then **zero** on a byte-identical cleaned-input prompt without a text repair. Route 2 changed from **eight to six flags** before substantive repair. Final audits could replay an accepted cache sample without reconciling the adverse verdict. These observations show audit nondeterminism, not measured error rates or evidence that the clean sample was correct. [R20, R22]

Round 6 (`d9ac619`) introduced a **k=3 union gate**: fresh requests bypass the response cache, and any current or valid prior same-prompt adverse flag remains binding. It is neither majority voting nor a claim of statistically independent samples. Repairs address the union and return through verification and fresh audit. The exact final-stage batches were: [R20]

Route / batch	Flags per sample	Prior adverse	Union	Unanimous / sentences	Verdict
1 / 1	0, 0, 36	44	47	209/245	reject
1 / 2	41, 52, 42	none	53	234/246	reject
1 / 3	0, 0, 0	none	0	245/245	accept
2 / 1	30, 0, 0	none	30	214/244	reject
2 / 2	2, 0, 5	none	6	238/244	reject
2 / 3	1, 15, 1	none	15	230/244	reject
2 / 4	0, 0, 0	none	0	244/244	accept

Unanimity counts agreement in binary votes; unanimous adverse votes still reject. Route 1 used **10 actual fresh audit calls** for **nine valid final-batch samples**; route 2 used **13 calls** for **12 samples**. Initial audits and bounded schema correction explain the difference. The first route-1 batch retained the earlier 44-flag result despite two clean samples. Complete valid adverse history therefore worked; the incomplete-history loophole later found by review required the repairs recorded in §7.5. [R20, R23]

Both round-6 lifecycles exported, but **each final bibliography had zero entries and each body had zero citation markers**. Native word counts were **5,387 / 4,413**, or **5,336 / 4,385** without machine markers. Both originality verdicts remained `possible_overlap`. External verification had removed the local source citations; subsequent clean audits did not restore them. These are native exports, not submission-ready essays or proof of source fidelity. Route 1 ran before a later cleanup-length/resume-budget correction; route 2 used the corrected source. The report preserves that source-version distinction rather than claiming both ran at the final committed head. [R20]

At the R6 checkpoint, citation repair was pending in R7; §7.3 gives the completed outcome. The requested fix verifies frozen-T references against local full text and page evidence first, uses web search as supplementary evidence, and refuses an export whose source claims lack citation coverage. It also must preserve valid adverse evidence from incomplete audit inventories. Publication's separate default floor is **3,000 words**, distinct from the historical writer construction target of **4,500** and the **6,000-word** ceiling. Native counters and marker-free counts must not be interchanged. [R20, R22, R23]

7.3 Citation retention repaired; R7 still refused

Nu4 R7 at `72c11e7` repaired citation retention and partial adverse-history handling. Each earlier route uses **two support commitments from one paper**, `gsv1996`; two bibliography entries are not two papers or evidence of a 26-paper synthesis. Citation-only restoration preserved the prose and restored two entries and two markers per candidate; the retained source spans were locally grounded at physical PDF page 18. Both candidates nevertheless refused. [R29]

Earlier route	Recorded sentence audit	Candidate bibliography / markers	Native words / marker-free words	Native export
1	141/245 unsupported; after the remaining repair, 173/246 unsupported	2 / 2	5,199 / 5,148	Refused
2	123/244 flags: 121 unsupported and two miscited	2 / 2	4,413 / 4,385	Refused

Successful R7 native exports: 0/2. Final k=3 samples: not run for either candidate. Both stopped before native citation/originality verification and final consensus. The local verifier diagnostic retained both entries, but did not create a native acceptance event. No favorable rerolls were used and stale exports were removed. R7 also corrects the earlier word-counter description: native words include title/body and markers, **exclude bibliography**; the comparison removes markers from that same scope. Restoring citations fixes a gate defect without making unsupported prose source-faithful. [R29]

7.4 All candidates, with final selection reserved

The operator authorized drafting every eligible candidate; the operator retains final submission selection. CLOSED routes and OPEN routes with an accepted inference enter the native queue. The later proposal-mode assignment separately covers selected root-only hypotheses under explicit labels; it is not permission to relabel them CLOSED-derived. [R33, R41]

Candidate	Status and time	Evidence and stopping point
gap-T2-analysis	PDF; rendered at the earlier v5 checkpoint	3,879 complete-Markdown words; 4,064 PDF-inclusive words; 17 references; 10 pages / two figures. Analysis of ARA output. No native typed audit, export or Opus pass. Its gap2-pending statement is historical; §2.5 supersedes it here.
gap1pu-B-sign	Refused, 03:14:26 UTC	4,790 words, five bibliography entries; successive audit flags 92 → 101 → 32 → 18. Final 15 miscited + three unsupported claims; missing node n1; three repairs used. Final k=3, citation/originality verification and polish not reached.
gap1pu-B-sign-short	Refused, 03:42:33 UTC	Writer failed the 2,500-word minimum. Attempt counts 1,412 / 1,373 / 0 / 1,203; 1,203 is the last rejected response, not a validated draft. Typed audit, final consensus, verification and polish not run.
T2, B/1, B/2, A/3 proposal-mode assignments	Pending, 06:28:53 UTC	Native implementation, draft, audit, export and polish results not yet credited.

The short retry's policy was an **orchestrator decision**, corrected in correspondence at **03:19:32 UTC**, not a separate human instruction or a competition minimum. For its three-node route with two direct source leaves from two papers, the policy used a 2,500 floor, 2,700 target, 3,000 requested limit and 6,000 runtime ceiling.

Audit samples, repair limit and citation policy stayed unchanged. Its failure does not establish that short essays are impossible; it records this bounded writer attempt. Neither refused candidate was enqueued for polish. [R42]

7.5 Audit history and citation verification must survive retries

R2-F1 exposed dropped partial adverse history on resume; R2-F2 exposed citation stripping. DLV-F1 then showed that valid adverse labels inside rejected schema retries of the current fresh batch could disappear when a corrected response was clean. Pi3's `3c9c8a4`, included in delivery round30b `6c6cf84`, unions those labels before verdict. The reviewed pipeline verifies frozen-T source evidence locally, restores only citation markers without changing prose, preserves semantic repair budgets, and independently checks support coverage before export. Missing local evidence is not positive mismatch evidence. Clean sampled judgments still do not prove source fidelity or statistical independence. [R23, R29, R31]

8. Runtime, prompts, human time and accounting

8.1 Providers and isolation

Role in the recorded work	Model / version evidence	Limit
API extraction and early construction	Receipts identify `claude-opus-5`; early cheaper-model experiments requested `claude-sonnet-5-5`	API ledger scope below excludes some early experiments. No underlying weight revision beyond reported identifiers is available.
Subscription construction/search	`claude-opus-5-5`, Claude Code `2.1.306`	Vendor runtime context remains; effective prompts differ from direct API prompts.
Codex pilots and writing	Requested `gpt-6-luna`, `codex-cli 0.163.0`	Serving identity was recorded as `unverified` when omitted by CLI output. A requested pin is not a serving receipt.
Publication-only polish	`claude-opus-5-5`, Claude Code `2.1.306`	Separately authorized polish, not authorization to resume Claude pilots.
Scaffold coding, orchestration, this report	Agent roles recorded in briefs/correspondence	A complete per-worker model/version and token ledger is unavailable in the supplied sources. No exact model identity is inferred from a role name.

The API-to-Claude switch followed the credit failure and the human request for an agent route. The Claude-to-Codex switch followed the operator’s clarification that “chat” meant ChatGPT/Codex; the active Claude pilot was halted. Later, the operator explicitly prohibited further API credit use and authorized ordinary agent calls by default, with Claude Opus reserved for the requested final polish. These were human choices, not automatic provider fallback. [R1, R13]

The Claude canary and review established the recorded configuration’s suppression of operator customizations and identity output. They did not establish absence of vendor built-ins or byte-identical API context. Subsequent reviews caught escaped-output leakage and unsafe diagnostic text; allowlisted diagnostics replaced free-text retention. Codex used fresh processes, temporary configuration, isolated host homes/process state, disabled project instructions, skills, memory, and external tools, plus a context canary. Generic built-in tool definitions could still be visible; actual tool events rejected the response. Strict-schema projection required its own

preflight. Isolation evidence is specific to the tested transport/version and threat model, not proof against all contamination. It does not erase pretrained model knowledge or establish that every generated idea came only from the supplied papers. [R13]

The later web-evidence transport deliberately differed from the earlier tool-free transport. Round 4 rejected an unidentified native event; round 5 diagnosed the specific web-search event and narrowly permitted it for `search_evidence`. Subsequent native exports are recorded in §7.2, together with their citation loss. That repair did not authorize other tool events. [R7, R20]

API construction stopped with opaque request failures, then a count-endpoint diagnostic identified insufficient credit. The earlier request failure was suspected to have the same cause, but that retrospective attribution was not proven. A subscription route was then implemented. Its first strict isolation probe failed because built-in context, hooks, and identity-related material remained. The revised contract excluded operator content and customizations while disclosing unavoidable vendor context. Output and decoded-string scans, fresh sessions, bounded retries, and canaries were reviewed repeatedly. [R1, R13]

Subscription capacity was also misestimated. A calibration comparison of aggregate usage with initial context inflated a factor from **0.553 to 1.424**, leaving **58 of 70** windows skipped in the first Claude pilot. Like-for-like accounting and a local rather than inherited token cap repaired the comparison. A later **401** after host authentication refresh exposed stale in-memory credentials; per-call credential reads and one bounded re-read retry addressed it. These repairs did not prove that future provider behavior would remain stable. [R1, R13]

8.2 Prompt disclosure

Stage prompts are topic-neutral templates combined at runtime with source text, the frozen catalogue, target/route state, explicit schemas, and correction feedback. Full prompt and response receipts were retained. Worker briefs separately specified implementation tasks, budgets, permitted providers, stop conditions, and expected evidence. They directed scaffold development; they were not all passed verbatim to philosophical generation calls. [R2, R3, R15]

Representative recorded instructions included:

- Extraction: “Preserve the paper’s argument structure” and “Do not reduce this to a fixed number of central claims.”
- Construction: a selected source inference must become a content conditional using the exact endpoint formulas, rather than a new standalone atom.
- Target proposal: assert a thesis in the system’s own voice; preserve frozen meanings; do not invent support to fill the requested pool.
- Search: instantiate the supplied bank and answer critical questions from available evidence. The engine retains licensing authority.
- Writing: compose from the selected route and source context, preserve citations and node coverage, and meet explicit length bounds.
- Publication: improve presentation without adding claims, preserve citation/marker sequences, then run the native audit again.

The quoted extraction sentences are short instruction excerpts, not literature quotations. The other entries are functional descriptions. `prompts.md` supplies the exact recorded instruction prefixes and their digests, including successive variants. Current source templates alone would be insufficient to reproduce earlier runs because

gates, correction messages, schemas, effective provider instructions, and runtime inputs changed. Exact raw prompts remain privately recoverable through the event hashes. [R15]

8.3 Scoped accounting

The principal ledger is `reports/overnight/fork6/accounting.json`. Its report covers **67 lineage runs** and deduplicates identical event receipts across copied prefixes and imports. We reaggregated its MODEL receipts by provider and stage and checked equality with its measured, unknown-at-reservation, and token totals. Per-run totals must not be added: many include the same extraction and freeze history. [R4]

This is a measured lineage subtotal, **not total campaign cost**. It excludes some early smoke/development work, later writer/publication branches, coding and review-agent conversations, the orchestration conversation, and production of this metapaper. The early smoke receipt separately reports **$7.059330**. The initial development record also retained unknown usage reservations. No reconciled global invoice or subscription fee allocation is present, so no all-in dollar total is asserted. [R1, R4]

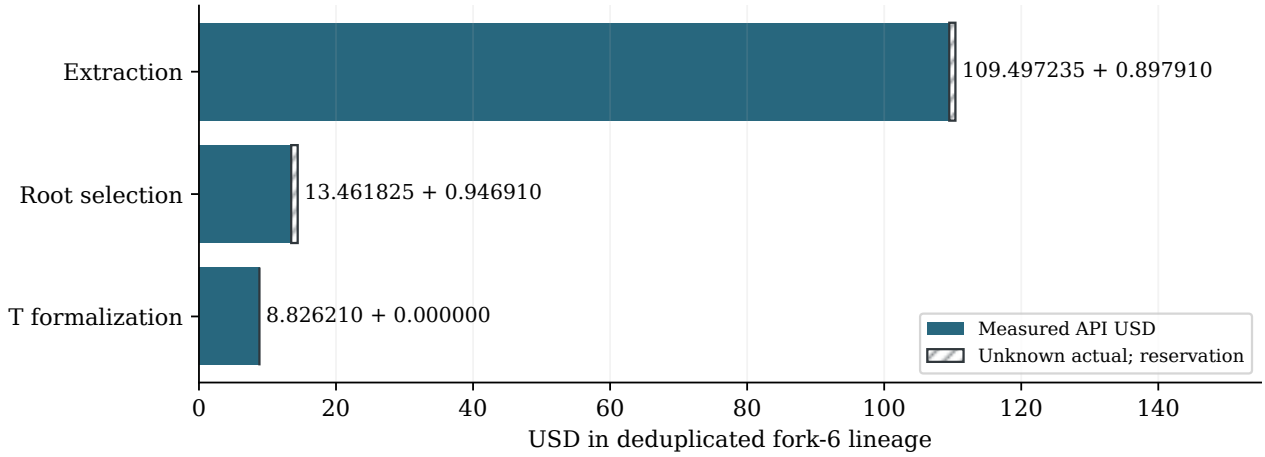


Figure 8. Measured API cost is $131.785270; unknown-at-reservation is $1.844820. The sum, $133.630090, is a budget-accounting figure, not an invoice. Subscription calls contribute no API USD here. Earlier development/smoke work and later writing/polish are outside this report's lineage.

The ledger contains **4,534 measured MODEL receipts** and **12 unknown-at-reservation receipts**. MODEL receipts include cache replays and validation/preflight work; these counts are not unique uncached generation calls. Recorded measured usage totals **113,530,631 tokens**, including the separate cache components. Unknown usage contributes **1,700,600 reserved tokens**, not measured consumption. API measured USD is **$131.785270**; the **$1.844820** unknown reservation makes the budget-accounted sum **$133.630090**. A failed response with no usage receipt does not become measured because it was budgeted. [R4]

Provider scope	Input	Output	Cache read	Cache write	Unknown reserved tokens
Anthropic API	11,314,239	3,008,563	0	0	175,424
Claude Code subscription	4,012	1,972,083	5,762,812	49,414,935	661,810
Codex CLI subscription	39,728,717	1,087,909	1,228,160	0	863,366
Legacy: provider absent	8,373	828	0	0	0

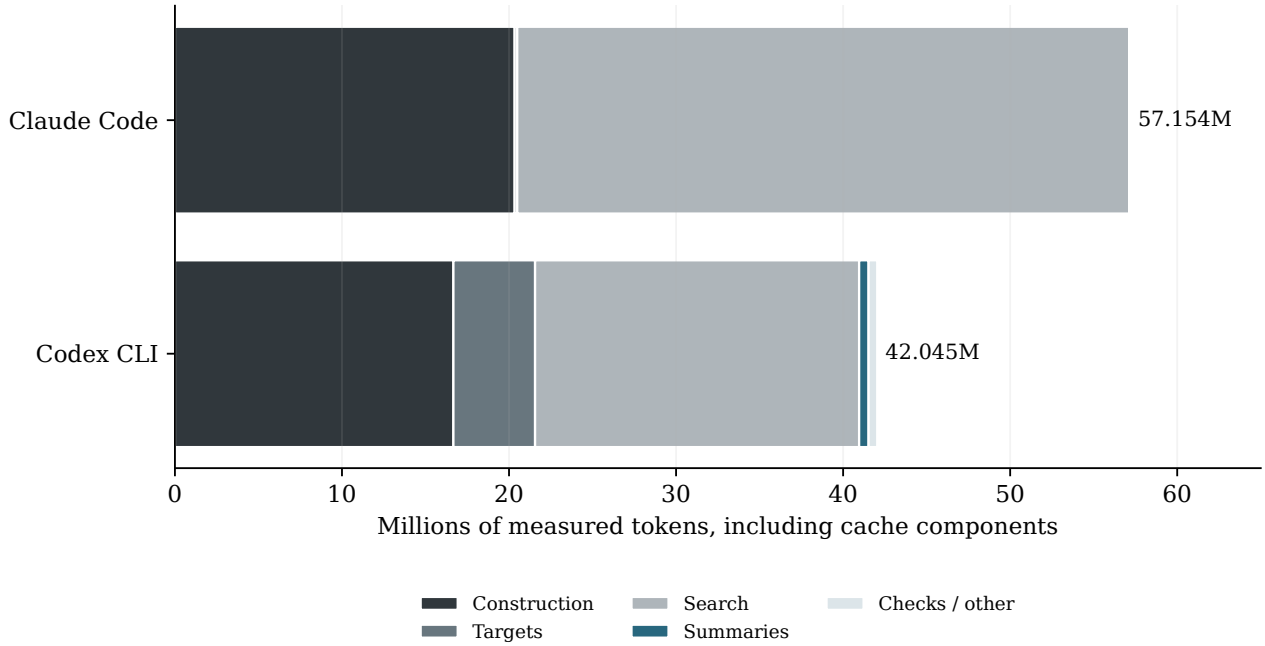


Figure 9. Measured subscription usage in the same deduplicated lineage, grouped by native stage. Cache reads and writes are retained; these are accounting components, not unique prose tokens. Unknown reservations and an unclassified-provider legacy receipt are shown in the accompanying table, not these bars. Coding-agent and orchestration conversations are outside this ledger.

A small legacy receipt requests `gpt-6-luna` but lacks an explicit provider field. Its **9,201 measured tokens** remain separately unattributed instead of being silently assigned to an API or subscription provider. Subscription list-price equivalents, where available, are not API payments. The data retains every native stage and token component; the figure groups stages only for readability.

Separate supplements and missing totals

The writer’s round-2 repair/audit supplement records **143,996 input**, **5,888 cache-read**, and **35,228 output tokens** across **eight actual responses**, all requesting Codex. These numbers exclude its preflight and earlier writing attempts. The publication receipt records **155,926 input**, **49,281 cache-read**, **208,242 cache-write**, and **183,617 output tokens**. Its **37 MODEL receipts** include cache replay; **19 dispatches** produced **18 measured responses** and one timeout with unknown usage retained at reservation. Thus its token aggregate is mixed measured/reserved, not wholly measured. These supplements are not added to the lineage subtotal. [R7, R8]

Gap pilot 1 has a separate local ledger: **242 fresh Codex calls**, comprising **221 main-run calls**, **20 schema-preflight calls**, and **one standalone canary**. Measured fresh usage is **4,769,322 input + 235,926 output + 266,752 cache-read = 5,272,000 tokens**; cache writes and server-tool uses are zero. New API spend is **$0**, while subscription dollar-equivalent is unavailable. Requested model identity remains `gpt-6-luna`; served identities are unverified. The pilot report’s inherited extraction charges are not new spend and are not added again to the fork-6 ledger. Its free recovery inspection made no new model calls. [R16]

Gap1r adds **197 fresh Codex calls**: **177 main-run calls** (two canaries, 115 alignment, 60 semantic classification) and **20 preflight calls**. Its measured components are **724,135 input + 36,474 output + 279,680 cache-read = 1,040,289 tokens**, with **$0 new API spend**. Together gap1 and gap1r used **439 calls / 6,312,289 measured token components**. Inherited extraction charges must not be added again. Gap1f and sigma r13's

offline plans made no new generation calls. At that earlier accounting checkpoint the enlarged-window A/B runs were pending; their later fresh usage appears below, and this report does not manufacture an R5–R6 global token total from their audit-call counts. [R18–R20]

Summed measured model wall time in the lineage is **99,802.960 seconds**, approximately **27.723 hours**; unknown-at-reservation receipts add **747.604 seconds** of recorded wall time. Calls can overlap, so neither sum is elapsed campaign duration or human labor. The extended correspondence span is **8 days, 21 hours, 7 minutes, 22 seconds**; the principal accounting and research checkpoints are unchanged. No complete operator-reported labor duration is available. Zero-minute fields on worker-generated command receipts cannot establish zero human effort. [R1, R4]

Fork	Fresh local Codex calls	Input tokens	Output tokens	Cache-read tokens
A	319	10,110,944	141,719	732,032
B	272	12,934,452	136,006	455,296
C	29	4,918,072	13,864	31,232

These components include local runtime canaries and reporting summaries, and exclude inherited extraction/construction and separate preflights. Shared A/B preflight and C preflight each used **20 calls**. Input/cache overlap is not assumed, and inherited charges must not be counted again. New API spend was **$0**; subscription dollar cost remains unknown. The earlier cost figures retain their fork-6 scope. [R28]

No complete campaign total for gap2, gap3, the wider forward sweep, new writer work or this report is inferred from the older lineage. Partial live counters are not final invoices. New API spend of zero in a Codex run does not mean free subscription usage or zero authoring cost. [R28, R35, R36]

8.4 Coordination and reporting errors

The coordinating agent acknowledged an idle period from **12:33:41 to 16:49:21 UTC on 28 October**, or **4 hours 15 minutes 40 seconds**. It waited for a go/no-go decision instead of starting an already available alternative in parallel. Earlier, its instruction to stop at “any cap” caused a worker to interrupt an extraction pass after a best-effort reconciliation refusal; that was a briefing error, not evidence that the corpus could not be processed. [R1]

Reporting itself needed repair. A **3.7 GB** HTML report crashed Chromium; a later sidecar-based report was approximately **2.45 MB**. Fork 6’s native report rendered on desktop but retained **19 pixels** of mobile overflow. Its event-origin index also mislabelled **15,663** inherited-prefix occurrences; a separate overlay corrected attribution without altering native report bytes. The main accounting deduplicates receipts, so imported lineage must not be summed as new spend. These defects concern the original run reports; this metapaper has its own rendering checks. [R1, R4]

The original all-candidates short-length attribution was corrected rather than counted as an operator action. Old report directories were archived only after verification according to the correspondence; source hashes in this paper identify snapshots, not a guarantee that every old source path remains unpacked. The operator's recent scheduling and parallel-work instructions are disclosed in Appendix A. [R1, R42]

9. Novelty, reviewed mechanisms and the remaining limits

9.1 Honest novelty status

Zero substantive CLOSED-derived contributions have been established so far. The measured paper-local atomic representation of T is the unifying explanation for the failed cross-paper synthesis (§§2.3, 5.3); sigma r17 is the pending repair. ARA did propose near-T2 hypotheses. That is a proposal-generation result, not a licensed argument, an exact match to the dossier's only not-found thesis, or priority in the literature. The exact T2 is eps2-proposed and externally tested in ARA. All A/B/C routes remained OPEN; the available B sign argument remained partial. The fallback analysis is a rendered analysis candidate; both native sign drafts refused. [R27, R28, R38, R42]

The formal novelty gate is weaker than philosophical novelty. A target may pass source-relative checks without being a useful, faithful, defended or literature-new contribution. Conversely, a hypothesis may be interesting while lacking a missing bridge premise; accepting that premise as H would establish at most the disclosed conditional derivation, if actually licensed. Proposal-mode prose must argue for that premise. No pending feature or larger search is counted as its own success. [R11, R34, R39–R41]

9.2 Review findings ledger: mechanism proof has a scope

Finding or reviewed change	Disposition at the named head	Limit
S12-F1: ID-free integration fixture drift	Closed on later integrated sigma12/xi line	Fixture repair does not certify source-alignment meaning.
NP-F1–3: imported screening, title audit/cache, duplicate occurrence permissions	Closed for publication at pi3 `a6a2f9b`	No promotion of a failed polish fallback to a successful edit.
R28-F1/F2: padding and inconsistent union	Closed by xi29 `0c4d472`, independently re-reviewed	Placeholder regressions and gate closure are not a new philosophical result.
S13-F1/F2: merged control, stage-window mismatch	Closed at sigma14 `7d19ccc`; merged `893985a`	Actual client capacity and semantic quality not proved.
R2-F1/F2: partial adverse history, citation stripping	Closed in reviewed delivery `6c6cf84`	R7 actual exports remained 0/2.
DLV-F1: fresh schema-retry adverse labels lost	Old `d24288b` blocked; pi3 `3c9c8a4` included in round30b closes it	Clean retries cannot erase valid adverse labels.
Sigma15 anchors and context-derived budgets	Kappa4 merge verdict; 1,460/1,464 full tests, 77 Sigma controls and 25 synthetic-merge controls	No faithful new T or live pilot success follows.
Xi31 integration at `f76c1b3`	1,673/1,677 full tests; all 820 controls; no fixes needed	Four opt-in tests skipped; offline compatibility, not live or submission readiness.

The delivery round30b review separately ran 253/254 tests, 170 controls and 33 probes. Do not aggregate tests across heads into a claimed full-suite result. Sigma15's later full review closes the pending label from the earlier v5 checkpoint; xi31 checked the actual merge and retained both decision histories. Sigma16, xi32 and proposal-mode completion are still pending at this revision's checkpoint. [R21–R24, R27, R30, R31, R43]

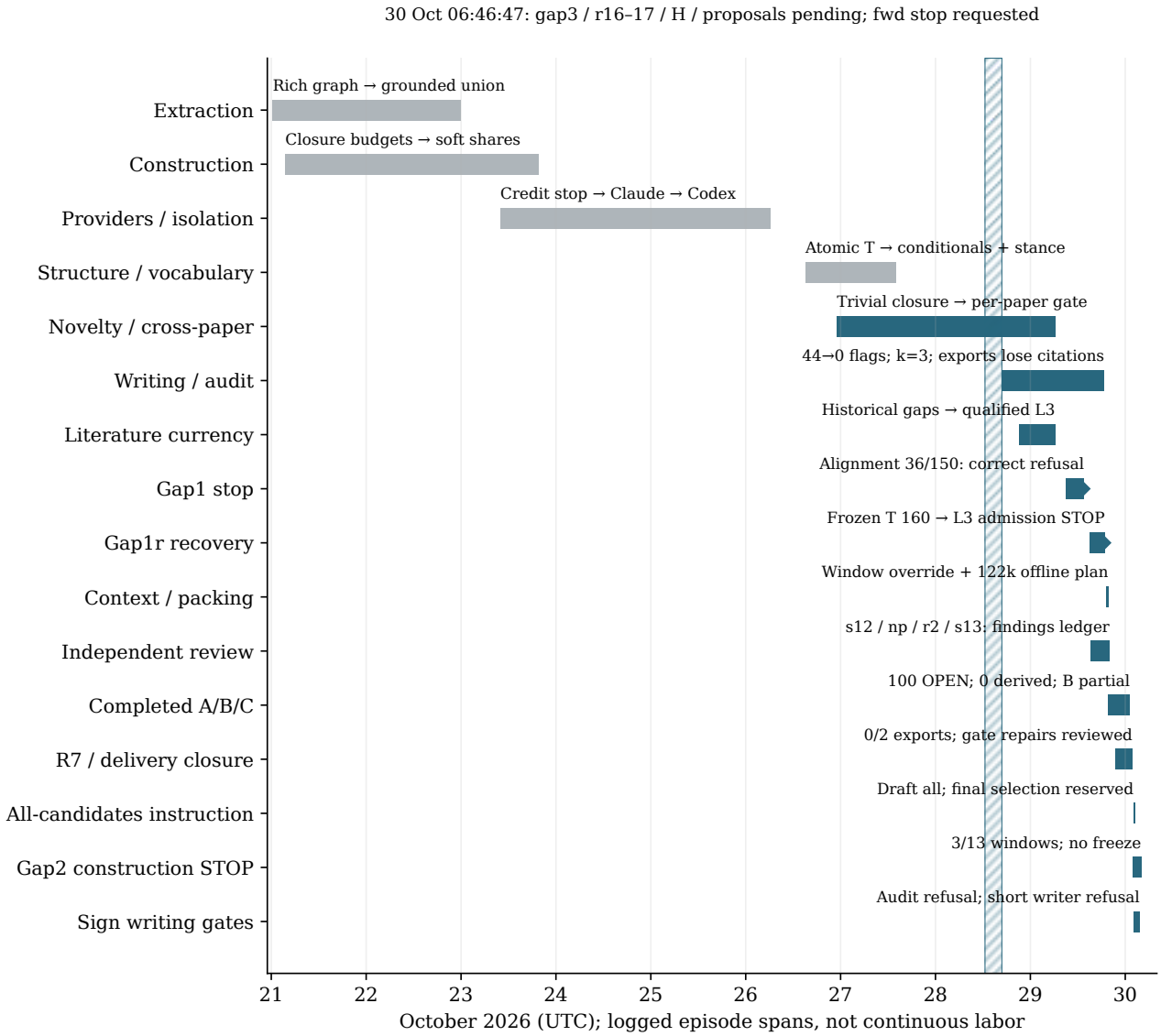


Figure 10. Logged iteration episodes. Diamonds mark gap1 alignment and gap1r target-admission stops. The hatched interval is the acknowledged coordinator idle period. Later bars record completed A/B/C and R7/delivery receipts; the all-candidates row records an instruction, not a research result. Historical packing plans preceded actual forks. Gap2 later stopped before freeze and both sign drafts refused. Gap3 and the new extensions remain pending; gate repairs do not establish semantic success.

9.3 Remaining limits and current pending register

Source selection remains incomplete and access-bounded; literal stance and source-fidelity requirements are separate from coverage. Atomic rules and a disputed alignment bridge remain material limits. Intake's prohibition on human philosophical content also has an unresolved literal tension with bibliographic titles and supplied filing labels; four such labels were not four selected theses. Corpus neutrality was not independently verified. [R2, R18, R28, R35, R36]

Privacy's early API refusal did not change visibility; the later operator report and coordinator check did. Anonymity is checked on these deliverables, not inferred for raw logs. Source acquisition, exact model serving identity, statistical independence of audits, full campaign cost and competition readiness are not certified. [R1, R13]

Pending item	Last evidence checkpoint, UTC on 30 October	What remains unproven
Gap3	06:22:42	Completed construction/coverage, alignment, freeze, usable anchor formulas, L3 binding, search and candidates.
Sigma16	06:44:42 report snapshot	Final implementation proof and review; parallel windows explicitly deferred.
Native forward sweep	06:46:15 stop instruction	20,000 census complete, zero leads and truncated; stop requested for the unfinished wider run, final stop/result receipt not credited.
Sigma17 shared vocabulary / wide alignment	06:46:47 instruction checkpoint	Reviewed shared-predicate formalization, wider alignment, measurable cross-paper applications and usable repaired T.
Xi32 hypothesis premises / planned gap4	06:28:53 instruction checkpoint	Reviewed native H support, one disclosed freeze, conditional routes and non-padding proofs.
Nu4 proposal mode and pi3 polish	06:28:53 instruction checkpoint	Native mode, labelled candidates, successful audit/export and polished PDFs.
Operator submission selection	06:28:53 instruction checkpoint	A final choice among delivered eligible candidates.

A pending item receives no completed result bar or invented success count. This paper's locally verified PDF is not a substitute for any of those gates. [R34, R36, R37, R39–R41]

10. Credits

The detailed historical rows below retain their recorded rounds and evidence. Their in-progress descriptions are dated to v4; the current updates in §10.6 and the concept sections supersede them. The new worker fwd brings the roster to **40 workers plus one orchestrator**. Agent IDs are not human names.

10.1 People, agent identifiers, and model evidence

The operator designed and directed the scaffold; AI coding agents built most of it under that direction. The operator chose the topic and literature, approved budgets and providers, set acceptance criteria and research priorities, selected writing routes, and requested publication and this report. Appendix A records the human actions. All names below are **agent identifiers, not people**. There are **40 distinct workers plus one orchestrator**, or **41 credited agent roles**. The count excludes the operator. [R1, R26]

Orchestrator — `brain` in correspondence, `@agent` in the live roster. The recorded harness is Claude and the model family is Claude; an exact model/version pin is unrecorded. From 21–29 October it planned work, wrote briefs, dispatched approved worker contexts, sequenced author/reviewer work, merged branches, coordinated budgets/providers, and reported to the operator. It also arranged output publication. These coordination credits do not transfer human design or research decisions to the agent. Evidence: `correspondence/log.txt`, `briefs/meta-credits.md`, and the read-only roster receipt in `data.json/credits`. Its acknowledged mistakes are retained: the 70–90-token construction estimate (§2.2), the “stop at any cap” briefing error and idle period (§8.4), and the logged delay in submitting the xi r29 prompt on 29 October. [R1, R26]

Each worker row records **Codex harness; exact worker model/version unrecorded**. The credits brief supplies the campaign-level Codex attribution. The initial log explicitly identifies alpha and beta as Codex; the v4 roster snapshot confirmed nine then-current worker contexts. For retired workers, that campaign attribution is not an

independently recovered per-session telemetry receipt. No worker is assigned `gpt-6-luna` merely because its code called that model. Model pins for ARA generation are listed separately in §8.1. [R1, R26]

The later Opus authorization concerned the **isolated publication polish pass** operated by pi3; pi3 itself is a Codex worker. “Opus for polish only” describes that later authorization, not the whole history: earlier API extraction and Claude subscription pilots used the models already disclosed in §8.1. No exact orchestrator subtype is inferred from the polish model. [R1, R8, R13, R26]

10.2 Intake, extraction, runs, and reporting foundations

Dates below are UTC calendar dates in October 2026 and identify recorded work/rounds, not continuous labor. A review credit names what the reviewer caught or verified; the corresponding author receives implementation credit. Report paths identify the evidence and its versioned qualifications. [R26]

Worker / harness / model	Recorded scope	Contribution, findings and evidence
alpha · Codex; model unrecorded	21 Oct; smoke series 5	Executed five live smoke runs at 5998c90 and recorded accounting/audits. The planned pilot was held, then reassigned to zeta; no alpha pilot result is credited. Evidence: `reports/alpha-live-runs.md`; `reports/alpha-pilot-section.md`
beta · Codex; model unrecorded	21 Oct; literature intake	Prepared intake for 53 bibliography entries; distinguished plain-text ingestion from full-text extraction. Flagged the committed-content rule tension and the lack of evidence for corpus neutrality. Evidence: `reports/beta-literature.md`
epsilon · Codex; model unrecorded	21 Oct; initial acquisition	Obtained 26 lawful PDFs from the 53-work list and recorded 27 access requests, versions, hashes and text-layer limits. No complete-access claim. Evidence: `reports/epsilon-pdfs.md`
gamma · Codex; model unrecorded	21 Oct; extraction rounds 1–6	Built per-paper extraction, rich graphs, bounded corrections, reconciliation and batched T construction. Changed the small-summary design after review; recorded Sonnet richness failures and repaired D4-B1 gap loss and D5-B1 continuation bounds. Final integration ba3a004. Evidence: `reports/gamma-extraction.md`; `reports/gamma-round6-report-draft.md`
delta · Codex; model unrecorded	21 Oct; initial review + re-reviews 1–6	Reviewed gamma. Identified missing assembled-context protection, cached lineage loss, the inadequate four-item cap, long-source/large-paper failures, reconciliation gap loss (D4-B1), and absent per-paper continuation bound (D5-B1). Cleared the final scoped repair at ba3a004. Evidence: `reports/delta-review.md`; `reports/delta-rereview6-section.md`
theta · Codex; model unrecorded	21 Oct; rounds 1–3	Repaired invalid context requests, added disclosed per-paper failure outcomes and completed-paper resume. Fixed iota T-B1 policy-sensitive reuse and T-S1; final be7468d. Evidence: `reports/theta-context-fix.md`
eta · Codex; model unrecorded	21 Oct; shared-T rounds 1–2	Implemented verified reuse of a single frozen T through --from-frozen, with source integrity and provenance checks. Rebased on theta and fixed E-B1/E-S1–3; final 185a318. Evidence: `reports/eta-shared-frozen-t.md`
iota · Codex; model unrecorded	21 Oct; theta/eta review and two re-reviews	Reproduced changed-policy resume bypass (T-B1), shared-T compatibility/provenance defects (E-B1/E-S1–3), and combined-suite failures. Rechecked both repaired branches before integration. Evidence: `reports/iota-review.md`; `reports/iota-theta-rereview-section.md`; `reports/iota-eta-rereview-section.md`

Worker / harness / model	Recorded scope	Contribution, findings and evidence
zeta · Codex; model unrecorded	21–29 Oct; full26, subscription/ Codex pilots, forks 1–6, gap1/r/f; A/B assigned	Operated extraction, pilots, frozen forks, route summaries and writing attempts; preserved failure/accounting receipts and performed source/substantive review. Corrected the coordinator's 70–90-token claim to about 233. Distinguished trivial/single-paper/padded results from contributions; documented gap1's ID stop, gap1r's 160-item freeze and admission stop, and gap1f's non-launch. gap1pu outcomes remain uncredited here. Evidence: `reports/zeta-full26-part7.md`; `reports/zeta-fork6.md`; `reports/zeta-gap1r.md`; `reports/overnight/gap1f/handoff.md`; `briefs/zeta-gap-fork2.md`
kappa · Codex; model unrecorded	21–22 Oct; acceptance rounds 1–3	Built real-response replay and extraction acceptance repairs, provider-schema preflight and bounded corrections. The live probe reached four of six papers, not full acceptance. Fixed lambda F1 diagnostic echo and F2 null-inventory recovery at 0eb8592; retained the deferred qualification. Evidence: `reports/kappa-acceptance.md`
lambda · Codex; model unrecorded	22 Oct; review + re-review	Caught request-content echo in diagnostics (F1) and null-inventory correction failure (F2), despite passing tests. Checked v4 reuse, required re-extraction of an incompatible paper, and confirmed the narrow repairs. Evidence: `reports/lambda-review.md`
mu · Codex; model unrecorded	22–23 Oct; rounds 1–7	Built range-union salvage, grounding/quarantine and monotone publication; disclosed placeholder output rather than calling it faithful extraction. Fixed snapshot/gap loss and pi's prerequisite defect, projected construction metadata, accelerated imports at 1d832b2, and added startup verification caching at cdc9753. Evidence: `reports/mu-four-papers.md`
omicron · Codex; model unrecorded	22 Oct; mu review + re-review	Caught accepted-inventory loss (B1), undisclosed legacy gaps (B2), and shared-T lineage incompatibility (B3). Reviewed mu/xi/eta integration and the repaired mu/xi rebase. Evidence: `reports/omicron-review.md`
pi · Codex; model unrecorded	22 Oct; mu round-4 review	Found union isolation omitted applies_to prerequisites (B1). Checked that the current published corpus had no affected instance; that bounded result supported the disclosed temporary integration, while mu repaired the general defect. Evidence: `reports/pi-review.md`
rho · Codex; model unrecorded	23 Oct; mu rounds 5–6 review	Verified prerequisite handling, metadata projection and import optimization. Caught a verification mismatch: the recount used an empty audience instead of the CLI default (S1). Required the pilot command to match the measured plan. Evidence: `reports/rho-review.md`
nu · Codex; model unrecorded	22 Oct; xi report/log review + re-review	Caught full-graph snapshot growth and repeated chain verification (B1), provenance defects (B2), renderer crashes (B3), and a remaining report failure (R1). Drove hashed sidecar delta tracing and honest legacy-ancestry disclosure. Evidence: `reports/nu-review.md`; `reports/nu-rereview-draft.md`

10.3 Construction, providers, search, and their reviewers

Worker / harness / model	Recorded scope	Contribution, findings and evidence
sigma · Codex; model unrecorded	23–29 Oct; construction rounds 1–14	Implemented bounded closure, global borrowing/soft shares, disclosed window skips and construction resume; diagnosed insufficient credit. Added linear formula validation, structured/stance-aware T, cross-paper alignment, ID-free replies and resume-alignment (r12, 4cefe35), then gap packing and explicit window forwarding (r13, c99c6b3). At the v4 checkpoint R14's S13-F1/F2 repair was in progress; §9.2 records its later closure. Evidence: `reports/sigma-bounded-closure.md`; `reports/sigma-round12.md`; `reports/sigma-round13.md`; `reports/sigma-round14.md`

Worker / harness / model	Recorded scope	Contribution, findings and evidence
tau · Codex; model unrecorded	23 Oct; bounded-closure review	Reviewed sigma's 5009af1 closure cuts, mandatory dependencies, omission attribution and global budget arithmetic. Confirmed the scoped implementation while retaining model-owned relevance and semantic limits; no blocking defect found. Evidence: `reports/tau-review.md`
upsilon · Codex; model unrecorded	23 Oct; sigma r2 review + r3 re-review	Found that nominally soft shares still vetoed closure admission and could starve windows (B1). Reproduced the repair at 1411e91 while checking global bounds and mandatory dependencies remained enforced. Evidence: `reports/upsilon-review.md`
phi · Codex; model unrecorded	23 Oct; sigma/xi r4 integration review	Verified construction-resume receipts/accounting and the compact report plus sidecars. Checked additive merge conflicts and offline resume; recommended integration without claiming a completed live pilot. Evidence: `reports/phi-review.md`
chi · Codex; model unrecorded	23 Oct; sigma r5 review	Reviewed count-endpoint refusal handling, explicit credit failure and plan-only resume at 0e0a7fc. Cleared the scoped code while preserving the operational credit block; did not infer a successful live recovery. Evidence: `reports/chi-review.md`
omega · Codex; model unrecorded	23–29 Oct; provider rounds 1–16	Implemented isolated Claude Code then Codex CLI transports, canaries, credential refresh, safe diagnostics, calibrated counting, local usage caps and strict-schema projection. Fixed psi/kappa2 provider findings. Later switched the configured default to codex_cli and tested new-paper extraction; r16 f7067a1 still failed both richness/structure trials under the repair-fraction gate. Evidence: `reports/omega-subagent-provider.md`
psi · Codex; model unrecorded	23 Oct; review + provider follow-ups/finals	Caught nondeterministic hashed-event ordering (F1), escaped-output leakage (F6), retry bypass (F7), credential transport exposure (F8), aggregate reservation misused as request size (F11), and lost correction-stage model override (F12). Rechecked the integrated repairs before the subscription pilot. Evidence: `reports/psi-review.md`
kappa2 · Codex; model unrecorded	24 Oct; omega/xi review + re-review	Found that sanitized diagnostic free text still retained synthetic credential values and short prompt echoes (F1). Reviewed failure continuation and stable publication order, then rechecked allowlisted diagnostics and integration f9c5a8f. Evidence: `reports/kappa2-review.md`
lambda2 · Codex; model unrecorded	24 Oct; auth/integration review	Checked per-call credential refresh, one bounded re-read retry, sanitization and accounting at omega 9a8e51a / integration 50ddc05. Found no blocking defect within the stated transport threat model. Evidence: `reports/lambda2-review.md`
rho2 · Codex; model unrecorded	25–26 Oct; OCR rounds 1–4	Integrated local PP-OCRv6_tiny, then PaddleOCR-VL-1.6 and span-level repairs. Reported 131/677 repaired spans and nine fully clean pages out of 82; did not claim complete recovery. Fixed tau2's confidence-dilution finding at 9aad013. These OCR models are task dependencies, not rho2's worker identity. Evidence: `reports/rho2-ocr.md`
tau2 · Codex; model unrecorded	26 Oct; provider/OCR/schema/resume reviews	Found OCR confidence dilution (F1) and stale awaiting_selection state that could suppress resumed search results (F2). Reviewed Codex isolation, strict-schema projection, linear formula validation, search continuation and their repairs; distinguished the actual pilot state from the general F2 defect. Evidence: `reports/tau2-review.md`
xi · Codex; model unrecorded	22–29 Oct; report/search/integration line, rounds 1–29; r30 assigned	Built ara report, sidecar traces, parallel search and resume-search; diagnosed goal/formula drift and atomic T. Implemented target/stance/novelty gates, CQ retrieval, semantic gap discovery and forward mining (r25–r29), integrating sigma. At 0c4d472 fixed R28-F1 Boolean padding, R28-F2 inconsistent-union suppression and S12-F1, plus solver lifetime. Delivery-branch r30 is an assignment, not a completed result here. Evidence: `reports/xi-report-tool.md`; `reports/xi-search-diagnosis.md`; `reports/xi-round25.md`; `reports/xi-round29.md`; `correspondence/log.txt`

Worker / harness / model	Recorded scope	Contribution, findings and evidence
upsilon2 · Codex; model unrecorded	26 Oct; structured-T/search integration review	Reviewed sigma r8 and xi r12; resolved the scratch integration at 6511ff6 and verified a native placeholder route closes while a flat negative stays open. Disclosed three merge conflicts and the need for a fresh live schema preflight. Evidence: `reports/upsilon2-review.md`
phi2 · Codex; model unrecorded	27 Oct; xi r13–r15 reviews	Reproduced single-commitment/tautology padding (B1) and attributed-report admission (B2), then checked their repairs. Found the single-support disclosure report crash (R1), later cleared at f89cd37. Evidence: `reports/phi2-review.md`
kappa3 · Codex; model unrecorded	27 Oct; sigma r9 / xi r16 integration review	Verified content definitions separated from attribution, stance-bound provenance and own-thesis/content classification. Produced the reviewed integration b48d56f; retained the distinction between offline validation and a useful real-corpus pilot. Evidence: `reports/kappa3-review.md`
rho3 · Codex; model unrecorded	27 Oct; sigma r10 / xi r17 integration review	Reviewed per-window construction correction and target-pool changes, exercised native defeasible closure, and committed scratch integration 7c7bdc8 with additive conflicts resolved. No blocking defect found in that scope; no live quality guarantee. Evidence: `reports/rho3-review.md`

10.4 Documentation, publication, literature, and methodology

Worker / harness / model	Recorded scope	Contribution, findings and evidence
omicron3 · Codex; model unrecorded	27 Oct; documentation cleanup + rebase	Reduced the README and organized concern guides, corrected stale/contradictory descriptions, preserved decision history and applied authorized identity redaction. Integrated xi's current fork/target guidance at 5b512de without executable changes. Evidence: `reports/omicron3-docs.md`
chi2 · Codex; model unrecorded	27 Oct; documentation review + re-review	Caught documentation integration conflicts and stale selection/fork guidance (F1). Checked command/link coverage, retained history and anonymity, then cleared the rebased documentation in the recorded re-review. Evidence: `reports/chi2-docs-review.md`
kappa4 · Codex; model unrecorded	27–29 Oct; xi r18–r29, sigma11–13, writer/publication reviews	Independently challenged novelty, stance, isolation and publication gates. Findings include R22-F1, R23-F1, R25-F1/F2, W1, S11-F1–3, INTEG-F1, R28-F1/F2, S12-F1, NP-F1–3, R2-F1/F2 and S13-F1/F2. Recorded closures separately; the later xi29 review rejects both exact gap1r padding leads. §9.2 carries the reviewed dispositions; §10.4 dates the earlier credit snapshot. Evidence: `reports/kappa4-review.md`; `reports/kappa4-writer-review.md`; `reports/kappa4-sigma11-review.md`; `reports/kappa4-r2-review.md`; `reports/kappa4-sigma13-review.md`; `reports/kappa4-xi29-review.md`
nu4 · Codex; model unrecorded	28–29 Oct; writer rounds 1–6; r7 in progress	Built sectioned composition and targeted revision, repaired sentence splitting and imported-draft screening (W1), and added typed audit roles. Fixed the exact web_search Other event in r5 (0368a17), exposed 44→0 flag variation, and implemented k=3 fresh union audits in r6 (d9ac619). Disclosed two citation-free exports; R7 citation/adverse-history repair remains uncredited as complete. Evidence: `reports/nu4-writer.md`; `briefs/nu4-round-7.md`

Worker / harness / model	Recorded scope	Contribution, findings and evidence
pi3 · Codex; model unrecorded	28–29 Oct; publication + quality passes 1–5	Built isolated Opus polish, typed post-polish audit/repairs, deterministic argument figures and native PDF rendering. Delivered 0aa6a29 and later quality passes; fixed NP-F1–3 at a6a2f9b. Q5 b6fc454 adds bounded structural retries and sampled publication audits, at the v4 checkpoint final R7 integration/live polish remained pending; a failed polish fallback is not a successful edit. Evidence: `reports/pi3-publication.md`; `reports/pi3-quality3.md`; `reports/pi3-quality5.md`; `reports/kappa4-r2-review.md`
eps2 · Codex; model unrecorded	28–29 Oct; modern literature, L3 and novelty dossier	Acquired lawful bridge/modern full texts, checked older open-problem claims against 2018–2026 literature and qualified L3 as partially anticipated. The later dossier maps nine analyst-generated theses: four anticipated, four partial, one not found within scope; seven further lawful full texts added. This is literature analysis, not an ARA-derived novel result. Evidence: `reports/eps2-bridge-literature.md`; `reports/eps2-L3.md`; `reports/eps2-methodology-supplement.md`; `reports/eps2-L3-novelty.md`
meta · Codex; model unrecorded	29 Oct; metapaper v1–v4	Assembled the methodology, human-action ledger, source-linked numerical data, review dispositions and deterministic figures; rendered PDFs and checked accounting, anonymity and layout. V3 preserves the evidence boundary; v4 adds this complete named-worker credit register. Evidence: `briefs/meta-paper.md`; `briefs/meta-credits.md`; `reports/papers/metapaper/v3/verification.json`

Later receipts, dated separately from the research checkpoint. The **20:42:29 UTC** correspondence entry records kappa4's xi29 review and integration of `0c4d472`. Its report closes **R28-F1, R28-F2 and S12-F1** and rejects the exact gap1r padding leads in offline reproduction. This is a later software-gate result; it does not rerun gap1r or create a substantive route. [R1, R27]

At **20:55:52 UTC**, eps2's L3 dossier superseded the earlier "dossier pending" label: **nine analyst-generated candidate theses**, **four anticipated, four partially anticipated, one not found within scope**, plus **seven** further lawful full texts. The not-found thesis is the exact necessity-and-sufficiency claim linking safety failure to obligatory retraction. "Not found" does not establish novelty, viability, priority or a successful ARA derivation. The dossier also reports that both previously inaccessible modern comparison papers were now obtained and inspected; other access limits remain. Its thesis map must not be presented as targets that zeta generated. [R1, R27]

At the v4 credit checkpoint, sigma14, nu4 R7, A/B/C and delivery integration were still pending. Their later completed receipts are now reported under construction/context (§§2–3), search (§4), writing (§7) and review (§9.2). New sigma16, xi32, forward-sweep and proposal-mode work remains pending at 06:28:53 UTC on 30 October. This preserves the difference between an assignment and an executed result. [R26–R43]

10.5 Roster reconciliation and limits

The original 39-worker roster was reconciled against the correspondence prefix and Markdown under both `briefs/` and `reports/`, including nested evidence/handoff directories. Generated metapaper copies were excluded to avoid counting this table as evidence for itself. Large native-run transcripts were hash-inventoried but excluded from worker-name discovery; model-generated text is not worker-authorship evidence. `data.json/credits` preserves the source inventory, hashes, candidate mentions, per-worker references and harness attribution limits. Review-only workers and earlier retired workers receive their own rows. This is a record-based attribution, not a recovered total of their tokens, elapsed work or individual philosophical authorship. [R26]

Two filenames would otherwise double-count workers: `briefs/omicron2-review.md` explicitly names **kappa2** in its title/body and required completion line; `briefs/sigma2-review.md` explicitly names **kappa4**. They are aliases in the record, not evidence of additional omicron2 or sigma2 contexts. Strings such as mu4/mu6, theta2/theta3, sigma5/sigma11/sigma12/sigma13 and xi29 refer to rounds; `eps*` is a family shorthand in the request. The distinct recorded literature workers are **epsilon** and **eps2**. No independent nu2, nu3 or pi2 context was found in the inspected records, so none is invented. [R26]

10.6 Current responsibilities, new worker, and retirement

Worker / harness / model	Recorded scope	Contribution, findings and evidence
fwd · Codex; model unrecorded	30 Oct; native forward sweep	Operator-approved sweep of cross-paper premise combinations on gap1r's authenticated freeze at `f76c1b3`; preserve licenses, SAT, stance and anti-padding gates; send qualifying routes for writing and later eps2 literature checking. Sweep completion not credited; the bounded census and subsequent stop request are recorded in §5.3. Evidence: `briefs/fwd-sweep.md`; `reports/fwd-sweep.md`; correspondence.

Omega retired from the active roster at 06:27:19 UTC after finishing the recorded provider work; the orchestrator retired its idle context when assigning fwd. This does not erase omega's contributions in §10.3 or imply that fwd inherited its context. **Sigma** supplied the reviewed r15 mechanisms, reports r16 under verification with concurrency deferred, and now prioritizes pending r17 shared vocabulary and wider alignment. **Kappa4** completed the r15 independent review; **xi** verified integrated `f76c1b3` without repairs and now owns pending r32. **Zeta** completed A/B/C, documented and stopped gap2 before freeze, and operates gap3. **Nu4** recorded both refused sign candidates and owns pending proposal-mode writing; **pi3** waits for eligible exports and must preserve proposal-origin labels during polish. **Eps2** supplied the bounded dossier and the exact external T2; future forward candidates still need its literature assessment. **Meta** produced the fallback analysis and unified this methodology around the same L/T/H boundaries. The **orchestrator** coordinated the pipeline and corrected its mistaken attribution of the short-length policy to the operator. [R1, R34–R43]

Appendix A. Complete recorded human-action entries

Timestamps are UTC. Entries are grouped exactly by correspondence record; one entry may contain several instructions or actions. H01–H03 are indirect reports of operator actions. Summaries remove identity and credential locations while preserving the decision. The corresponding anonymised record text and original line number are in `data.json`.

ID / UTC timestamp	Recorded human action
H01 · 2026-10-21 00:02:03	Approved the initial coding workers and spending: five smoke runs capped at $5 each and one pilot capped at $100. This approval is reported by the coordinator.
H02 · 2026-10-21 00:12:18	Chose per-paper logged extraction; the existing pilot was held pending that capability.
H03 · 2026-10-21 00:48:04	Asked the team to collect PDFs; the operational brief restricted collection to lawful open access.
H04 · 2026-10-21 02:03:18	Chose rich extraction rather than the compressed-summary option and said they were obtaining remaining papers.
H05 · 2026-10-21 03:38:41	Approved one additional build-T attempt by increasing its cap from $4 to $6.
H06 · 2026-10-21 03:40:20	Directed extraction to use a cheaper model and asked the team to get through to the pilot.
H07 · 2026-10-21 18:21:02	Pulled and pushed the main branch to the remote repository.
H08 · 2026-10-21 18:25:51	Approved an approximately $600–700 all-in budget, a shared frozen T across candidates, and the start of extraction/pilot work.
H09 · 2026-10-22 03:32:13	Asked for immediate repair of the four-paper failure policy; requested looser workable restrictions and OCR if needed.
H10 · 2026-10-22 05:33:48	Requested a full overnight pass, a $200 cap, and a detailed morning report with search-tree traces; authorized workers and method iteration. The coordinator interpreted the cap as new spend within the earlier campaign budget.
H11 · 2026-10-23 13:17:07	Requested an agent/connector route, a fresh T rebuild, parallel work, and iteration until an acceptable pilot.
H12 · 2026-10-24 15:28:25	Clarified that “use chat” meant ChatGPT/Codex and prohibited Claude-subscription pilots. The active pilot was halted.
H13 · 2026-10-25 22:26:45	Requested integration of the new open-source Baidu OCR family in parallel for future content recovery.
H14 · 2026-10-25 22:53:20	Completed a separate ChatGPT/Codex login. No credential value or identifying location is reproduced.
H15 · 2026-10-27 02:06:18	Requested plain-language documentation and repository organization using an existing internal methodology; the resulting assignment was documentation-only.
H16 · 2026-10-27 15:15:37	Saved a repository access token. The coordinator then pushed and started periodic pushes; no credential value is reproduced.
H17 · 2026-10-27 15:38:31	Approved the complete pilot and entry criteria, including 20/26-paper coverage, three qualifying derived routes, a 4,500–6,000-word audited essay, an anonymised methodology, and the October 31 AoE deadline. Set October 29/30 checkpoints.

ID / UTC timestamp	Recorded human action
H18 · 2026-10-28 16:52:30	Directed “Write C,” asked for at least one resolvable literature gap, and asked to build all routes. The coordinator recorded selection of all available CLOSED-derived routes for writing.
H19 · 2026-10-28 20:59:16	Requested a Claude Opus final pass producing professional papers, visualizations, Markdown, then PDF.
H20 · 2026-10-28 21:02:01	Questioned the lack of novel conclusions; requested bridge-paper discovery/ingestion, creative investigation, consideration of architectural changes, and examination of contradictions.
H21 · 2026-10-28 21:03:17	Required explicit disclosure of missing modern literature, acquisition of that literature, and accurate claims about open questions.
H22 · 2026-10-28 21:03:45	Directed the team to obtain the full texts and put them in the full-text collection.
H23 · 2026-10-28 21:06:14	Requested maximum parallelization of the ongoing work.
H24 · 2026-10-28 21:14:27	Prohibited further credit-based API work and authorized normal agent calls as the default; the coordinator retained Claude only for the separately requested Opus polish.
H25 · 2026-10-29 06:17:05	Gave GO for the safety–retraction gap paper and this visual metapaper; approved “not found within stated search scope, partially anticipated,” directed best effort from available sources, and requested an output repository.
H26 · 2026-10-29 06:20:26	Asked to make the scaffold repository private and place outputs in a folder for later deletion. The visibility API request was refused with HTTP 403 for insufficient administration permission; that action did not change visibility; H28 records the later successful change.
H27 · 2026-10-29 19:09:07	Authorized loosening the token limit. The later adapter change forwards an explicit provider window; raising native admission alone was insufficient.
H28 · 2026-10-29 20:04:19	Reported the repository private; the coordinator verified visibility through the API, copied outputs, and pushed them. This supersedes the earlier failed privacy request.
H29 · 2026-10-29 20:41:14	Questioned idle completed agents and context limits, and asked whether the work was pursuing real novelty. The coordinator requested an L3 novelty dossier, delivery integration, and this report revision.
H30 · 2026-10-29 20:56:46	Requested a repository push. The coordinator recorded main and four work-in-progress branches as pushed; this was a publication action, not a research-result approval.
H31 · 2026-10-29 21:09:25	Requested named credits for every worker in the metapaper. Worker identifiers refer to agents; the human remains the operator.
H32 · 2026-10-30 02:09:23	Authorized drafting every available candidate as soon as it is ready; final submission selection remains with the operator. The coordinator assigned native writing/audit to nu4, polish to pi3, and the separate fallback analysis to meta.
H33 · 2026-10-30 05:47:40	Reported apparently empty pilot process windows. The coordinator explained that ARA logs to files, closed completed A/B windows after retaining results, and confirmed gap3 remained live.
H34 · 2026-10-30 05:51:57	Approved construction improvements while requiring the current run to stay on schedule; the coordinator assigned default-off repairs without touching gap3.
H35 · 2026-10-30 06:22:01	Requested parallel work and the maximum number of papers, and expressed that one output seemed novel. The coordinator distinguished near-T2 proposals from derived results and assigned four proposal-mode candidates plus safe parallel construction work.
H36 · 2026-10-30 06:27:19	Authorized proposal papers and hypothesis-premise work in parallel without stopping ongoing work; asked whether premise combinations and routes were being used to find literature gaps. The coordinator disclosed bounded earlier forward coverage, retired idle omega and assigned the native fwd sweep.

ID / UTC timestamp	Recorded human action
H37 · 2026-10-30 06:28:39	Approved the proposed hypothesis-premise extension; the coordinator assigned xi round32.
H38 · 2026-10-30 06:28:53	Requested an immediate metapaper update unified under ARA base concepts.

Earlier human actions without a timestamp in this log. The human designed and directed the scaffold, chose the topic, and supplied/committed the initial bibliography and filing labels. The intake review found no added argument annotations, but flagged an unresolved literal-rule tension: bibliographic titles and human-provided grouping labels can name positions despite the repository’s blanket prohibition on committed philosophical content. Four position-related filing labels were not four selected theses. The collection’s claimed completeness and neutrality were not independently verified. No selection policy is promoted here into proof that the corpus was balanced. [R2]

Current revision instruction without a separate timestamped operator entry. The operator directed meta to make the measured 141/160 atomic commitments, 2/147 cross-paper symbols and 150-pair alignment the central construction/forward-search finding, add a deterministic two-panel figure, and mark sigma r17's shared-catalogue/wider-alignment repair pending. The correspondence records the coordinator's diagnosis and dispatch at 06:46:15–06:46:47; it does not timestamp this operator message separately. This direction is disclosed without inventing an additional timed human-action row.

Appendix B. Evidence register and reproducibility

Each citation identifies a source snapshot, not a live guarantee that the file has not changed. `data.json/unified_source_snapshots` records this revision's new hashes and collection times; older hash sets retain their original scopes. Numerical statements are indexed in `data.json/numerical_claim_index`. Pending statuses are anchored to the stated times, even if work continues during rendering.

ID	Primary sources and responsibility
R1	`correspondence/log.txt`, frozen through 2026-10-30 06:46:47 UTC: decisions, iterations, human actions, and timing.
R2	`briefs/beta-literature.md`, `reports/beta-literature.md`, `reports/epsilon-pdfs.md`: supplied bibliography, intake scope, rule issues, initial access.
R3	Scaffold `CLAUDE.md`, `PLAN.md`, `SOURCE_MAP.md`, `DECISIONS.md`, and `docs/`: architecture, boundaries, lifecycle, policies and their limitations.
R4	`reports/overnight/fork6/{data,accounting}.json`, its content-addressed event sidecars, and `reports/zeta-fork6.md`: deduplicated accounting, gap/search outcomes and report defects.
R5	`reports/eps2-bridge-literature.md`, `eps2-L3.md`, `eps2-methodology-supplement.md` and their JSON twins: initial sweep, focused currency check, cumulative coverage.
R6	`reports/zeta-codex-pilot-5.md` and `reports/overnight/codex5/`: frozen structure, stance, symbol reuse, targets and routes.
R7	`reports/nu4-writer.md`, round-2 and round-4 native receipts, plus `zeta-writing-test.md` and `zeta-writing-route2.md`: length repair, audit variation, blocked retrieval and export.
R8	`reports/pi3-publication.md` and `pi3/runs/pi3-proof/live-result.json`: preserved input, polish, native audit, usage and rendering evidence.
R9	`reports/zeta-fork1.md` through `zeta-fork6.md`, native target-gate receipts and `kappa4-review.md`: rejection mechanisms, counterexamples, false positives and substantive route review.
R10	`reports/zeta-codex-pilot-2-resumed.md`, `zeta-codex-pilot-3.md`, `xi-search-diagnosis.md`: atomic T, conclusion mismatch and trivial closure.
R11	`briefs/xi-round-27.md`, `reports/xi-round27.md`: per-paper novelty correction, superseded census and final bounded zero-candidate census.
R12	`reports/zeta-full26-part7.md`, `gamma-extraction.md`, `mu-four-papers.md`, `sigma-bounded-closure.md`: extraction inventory, quarantine, retention and construction repairs.
R13	`reports/omega-subagent-provider.md`, provider review reports and scaffold `docs/providers.md`: API/CLI versions, isolation scope, calibration and diagnostics.
R14	`reports/zeta-conflicts.md`: bounded semantic-conflict pass and source/currency review.
R15	`briefs/` and run MODEL receipts; exact instruction variants in the prompt supplement: task direction, stage instructions, correction history and prompt identities. The instruction supplement remains the version-1 inspected scope.
R16	`reports/overnight/gap1/handoff.md`, `reports/zeta-gap1.md`, `pilot-exit.json`, `final-local-accounting.json`, and the preserved integration review: failed alignment, construction inventory, source limits, timing and fresh usage.

ID	Primary sources and responsibility
R17	`briefs/sigma-round-12.md` and `briefs/nu4-round-5.md`: historical repair requests, pending at v2; later results are in R18 and R20.
R18	`reports/overnight/gap1r/handoff.md`, `reports/zeta-gap1r.md`, `reports/overnight/gap1r/final-local-accounting.json`, `reports/overnight/gap1r/pilot-exit.json`: Gap1r recovery, frozen inventory, source/padding review, admission refusal, timing and fresh accounting.
R19	`reports/overnight/gap1f/handoff.md`, `reports/sigma-round13.md`, `briefs/zeta-gap-fork2.md`: Gap1f client-window inspection, sigma r13 offline plans/window plumbing, historical packed/unpacked fork brief; completed results in R28.
R20	`reports/nu4-writer.md`, `briefs/nu4-round-7.md`: Nu4 R5–R6 native exports, audit dissent and union batches, citation loss; historical R7 repair brief, completed outcome in R29.
R21	`reports/kappa4-sigma12-review.md`: Kappa4 s12: ID-free response and recovery review, integration fixture blocker.
R22	`reports/kappa4-nu4pi3-review.md`: Kappa4 np: publication screening/title/occurrence blockers, audit and length-policy notes.
R23	`reports/kappa4-r2-review.md`: Kappa4 r2: Pi closure evidence, incomplete adverse-history and citation-stripping blockers.
R24	`reports/kappa4-sigma13-review.md`: Kappa4 s13: stage-window admission mismatch, merged mutation-anchor failure, S12-F1 closure.
R25	`briefs/eps2-L3-novelty.md`: L3 novelty-dossier brief; historical request; completed bounded assessment in R27.
R26	`briefs/meta-credits.md`; per-worker report paths in §10; full inventory and read-only harness roster in `data.json/credits`: complete credits, aliases and model-attribution limits.
R27	`reports/kappa4-xi29-review.md`, `reports/eps2-L3-novelty.md`, `reports/sigma-round14.md`, `reports/pi3-quality5.md`: later completed receipts versus in-progress work, individually qualified in §10 (historical rows).
R28	Completed A/B/C fork handoffs: outcomes, source limits, context and fresh accounting. Sources: `reports/overnight/gap1pu/handoff.md`, `reports/overnight/gap1pu/A-handoff.md`, `reports/overnight/gap1pu/B-handoff.md`, `reports/overnight/gap1pu/C-handoff.md`
R29	Nu4 writer report, Round 7: citation/history repair, audit refusals and corrected word-count scope. Sources: `reports/nu4-writer.md`
R30	Kappa4 sigma14 review: S13-F1/F2 closure on the pinned head. Sources: `reports/kappa4-sigma14-review.md`
R31	Kappa4 delivery review, xi round30 and pi3 quality5: R2 closures and DLV-F1 closure boundary. Sources: `reports/kappa4-delivery-review.md`, `reports/xi-round30.md`, `reports/pi3-quality5.md`
R32	Gap2 live checkpoint and grounded-root review; sigma15 review was in progress at the earlier snapshot; R43 supersedes it. Sources: `reports/overnight/gap2/live-status.md`, `reports/overnight/gap2/source-anchor-review.md`, `reports/overnight/gap2/exact-root-prelaunch-check.json`, `reports/kappa4-sigma15-review.md`
R33	All-candidates brief and queue; fallback analysis source, status and verification. Sources: `briefs/all-candidates.md`, `reports/candidates/queue.jsonl`, `reports/candidates/gap-T2-analysis/paper.md`, `reports/candidates/gap-T2-analysis/status.json`, `reports/candidates/gap-T2-analysis/verification.json`
R34	`briefs/meta-v5.md`, `correspondence/log.txt`

ID	Primary sources and responsibility
R35	`reports/overnight/gap2/pre-freeze-check.md`, `reports/overnight/gap2/construction-outcome.md`, `reports/overnight/gap2/live-status.md`, `scaffold/runs/codex-gap-2-control/pre-freeze-operator-check.json`
R36	`reports/overnight/gap3/live-status.md`, `reports/overnight/gap3/work-notes.md`, `reports/overnight/gap3/substantive-construction-review.md`
R37	`briefs/sigma-round-16.md`
R38	`reports/overnight/gap1pu/l3-dossier-mapping.md`, `reports/overnight/gap1pu/l3-dossier-mapping.json`
R39	`briefs/fwd-sweep.md`
R40	`briefs/xi-round-32.md`
R41	`briefs/nu4-proposal-mode.md`
R42	`reports/candidates/gap1pu-B-sign/status.json`, `reports/candidates/gap1pu-B-sign-short/status.json`, `reports/candidates/gap-T2-analysis/status.json`
R43	`reports/kappa4-sigma15-review.md`, `reports/xi-round31.md`
R44	`reports/fwd-sweep.md`, `reports/fwd-sweep-wide/summary.json`: completed truncated census inside an unfinished sweep.
R45	`reports/sigma-round16.md`: local implementation checkpoint; final verification/review pending, concurrent windows deferred.
R46	`scaffold/runs/codex-gap-1r/frozen-t.json`, `scaffold/runs/codex-gap-1r/commitment-provenance.json`: independently counted syntax, catalogue reuse and alignment; method in `data.json/vocabulary_bottleneck`.
R47	`briefs/sigma-round-17.md`; correspondence at 06:46:15–06:46:47: shared-vocabulary repair assignment and wider-sweep stop request.

This report used no new ARA generation or Opus polish calls. Authoring-agent usage is outside the supplied native ledger. Deterministic SVGs are generated from recorded data; local Chromium renders the Markdown/HTML with browser networking blocked. Checks cover source and citation inventories, figure determinism, fonts, captions, page bounds, table rows, anonymity across delivered text and PDF metadata, and preservation of previous deliveries. These checks do not establish an audit pass or philosophical truth. The earlier v5 snapshot is preserved as `v5-checkpoint-0229/`; it is not the current paper.

Finding placement. The following ledger maps every material finding from the earlier working record and this revision to the concept section carrying it. Historical failure labels describe what failed; the referenced section gives the current disposition.

Finding	Report section
1. No substantive novel contribution or qualifying pilot	9.1
2. Historical R4 audit counts versus marker-free words	7.1
3. Gap1 correct ID refusal and missing continuation	2.4
4. Recovery and zero unresolved do not certify bridge semantics	2.4, 9.3

Finding	Report section
5. Source stance, citation spans, omissions and incomplete L3 anchors	2.2–2.6
6. SAT freeze excludes R28-F2 here; observed padding remains	4.2
7. Full target request refused despite smaller readiness check	3.2
8. Pinned window metadata and omitted adapter forwarding	3.2
9. Offline one-hop plan and actual two-hop fork A are separate evidence	3.3
10. Audit nondeterminism and k=3 union limits	7.2
11. Native exports lost all citations; R7 repaired citations but refused both exports	7.2–7.3
12. Incomplete adverse history can allow unchanged export	7.5
13. L3 partial anticipation, access limits and bounded completed dossier	5.1–5.2
14. Isolation and unverified serving identity	8.1, 9.3
15. Accounting and human record limits	1, 8, Appendix A
16. Target and native report losses	4.1, 8.4
17. Privacy failure superseded by later verified status	9.3, Appendix A
18. Intake ambiguity and unverified collection neutrality	9.3, Appendix A
19. S12-F1 fixture failure closed at later integrated head	9.2
20. NP-F1 imported publication screening closed for publication	9.2
21. NP-F2 title audit/cache omission closed for publication	9.2
22. NP-F3 duplicate occurrence permission closed for publication	9.2
23. S13-F2 stage-window over-admission closed at later reviewed head	3.2, 9.2
24. S13-F1 merged mutation control still closed at later reviewed head	9.2
25. Green scoped tests did not exclude review blockers	9.2
26. Separate length policies and source versions	7.2–7.4
27. Named-worker roster, aliases and uncertain model identities	10
28. Later xi29 review closes padding/union findings without a new pilot result	4.2, 9.2
29. Completed literature dossier is bounded analysis, not a native novel contribution	5.2
30. Current assignments and draft repair reports do not establish completed results	9.3, 10.6
31. Completed A/B/C: 100 OPEN, one partial, zero derived/substantive	4.3
32. Same freeze, 159/160 stance failure and 21 skipped windows remain	2.4, 4.3
33. External T2 atoms are not decomposed or proven false	4.3, 5.2
34. Missing sufficiency and necessity/exhaustiveness premises	6.1

Finding	Report section
35. Actual context counts, pre-packing disagreement and experiment confounds	3.3
36. Conservative admission is not actual serving capacity; recovery disclosed	3.3
37. Fork-local usage excludes inherited/preflight costs; subscription dollars unknown	8.3
38. R7 restores citations but refuses both; final k=3 not reached	7.3
39. Native word counter excludes bibliography; two entries from one paper	7.3
40. S13/R2 closures and DLV-F1 boundary are exact-head offline evidence	7.5, 9.2
41. Gap2 stopped and sigma15 reviewed; plans/anchors are not results	2.5–2.6, 9.2
42. Fallback analysis is rendered, not native-audited/exported/polished	7.4
43. Zero substantive novelty; bounded dossier is weaker than priority proof	9.1
44. All-candidates authorization preserves operator selection and worker credit	7.4, 10.6, Appendix A
45. Gap2 stopped before freeze: 3/13 accepted, 57 commitments/3 papers; no context refusals	2.5
46. Changed batch granularity/cache identity; no cut closure or plan exclusion	2.5
47. Gap2 gained atomic rule but lost safety; correction diagnostic still 94/95	2.5
48. Gap3 startup/allocation recovery and live counts are not completed coverage	2.6
49. Gap3 anchors atomic; stance and Mighty Failure paraphrase remain qualified	2.6
50. Sigma16 partial/split/attribution/parallel mechanisms pending, running gap3 protected	2.6, 9.3
51. ARA proposed B1/B2 near T2; exact T2 came from eps2 and mappings are partial	5.2
52. Forward coverage was bounded; wider native sweep pending	5.3
53. Hypothesis premise P is proposed support; conditional derivation would not prove P	6.2
54. Proposal-mode origin labels and audits remain required; four assignments not exports	6.3, 7.4
55. Sign audit refusal and short writer refusal; length policy was orchestrator choice	7.4, Appendix A
56. Sigma15 review and xi31 integrated verification completed but offline; fwd added and omega retired	9.2, 10.6
57. Forward census saturates on Boolean enumeration; configured depth is not coverage; final sweep pending	5.3
58. Sigma16 author checkpoint: complexity guard repaired; final proof pending, concurrency deferred and retry coverage distinct	2.6
59. Central T bottleneck: 141/160 bare atoms, 19 syntactically structured formulas, 147 zero-arity symbols	2.3, 5.3
60. Two shared symbols arise only through one bridge; zero direct cross-paper reuse; 149/150 alignment pairs unrelated	2.3
61. Sparse vocabulary unifies bounded padding/OPEN outcomes; sigma17 pending, wider forward stop requested, no impossibility theorem	2.3, 5.3, 9.1

61 findings, 61 placed.